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MODULE A

STATISTICS

Unit 1. Definition of Statistics, Importance & Limitations & Data Collection, Classification & Tabulation

Unit 2. Sampling Techniques

Unit 3. Measures of Central Tendency & Dispersion, Skewness, Kurtosis

Unit 4. Correlation and Regression

Linked

Unit 5. Time Series

Unit 6. Theory of Probability

Unit 7. Estimation

Unit 8. Linear Programming

Unit 9. Simulation

Linked

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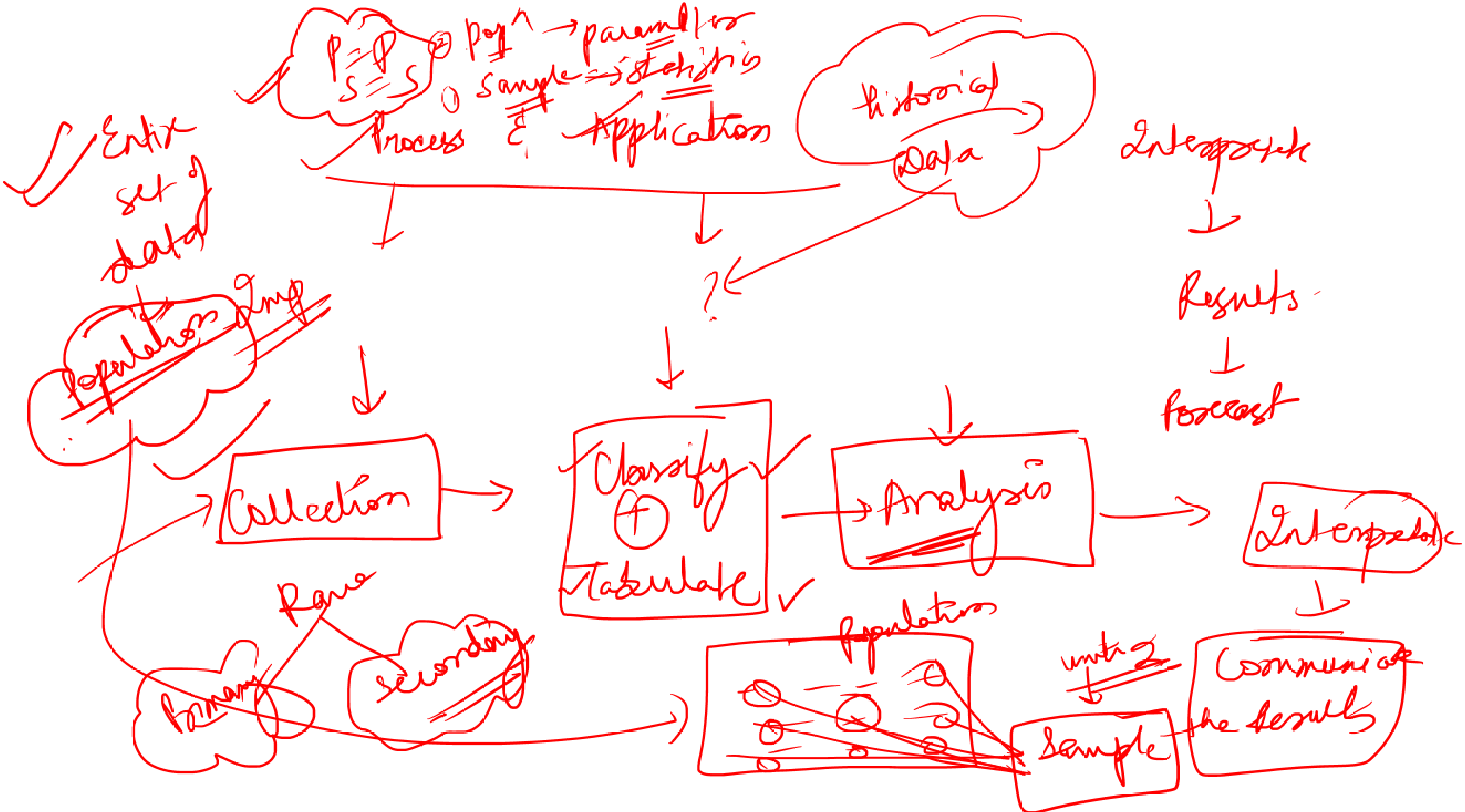
UNIT 1

Definition of Statistics, Importance & Limitations & Data Collection, Classification & Tabulation

STRUCTURE



P = P
S = S
pop^n → parameter
sample → statistics
Process & Application
historical Data
Interpret
Results
Forecast
Entire set of data
Population Imp
Collection
Classify + Tabulate
Analysis
Interpolate
Communicate the results
Raw
Primary
Secondary
Population
Sample

Class Int

0-10
10-20
20 30

Exclusive Series

9.5

0-9
10-19
20-29

9.5
to
19.5

Inclusive

CM = Mid = $\frac{L.L + U.L}{2}$

RF = $\frac{C.F}{Total\ F}$

% = $\frac{C.F}{T.F} \times 100$

RF × 100

→ U.L & LL

Continuous Class

F Dept or Density = $\frac{C.F}{Width}$

−LL = .5 +UL

UNIT 1 – Definition of Statistics, Importance, Data Collection, Classification & Tabulation

- **1. Meaning of Statistics**
- Statistics is the science of:
- Collecting data – collection
- Organizing data – classification
- Presenting data – tabulation
- Analyzing data –
- Interpreting data –
- It helps management and bankers take decisions scientifically.

2. Functions / Importance of Statistics

- Statistics is important in:
- **Banking**
- Credit appraisal
- Risk analysis
- NPA prediction
- Forecasting deposits & advances

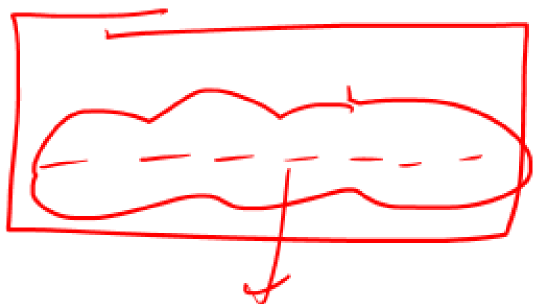
- **Economics**
- GDP measurement
- Inflation calculation
- Employment analysis
- **Business**
- Demand forecasting
- Inventory control
- Sales trend analysis
- **Government**
- Census
- Budget preparation
- Policy making

Census → Entire set of data
pop^n

Sample → Representation of pop^n

Limitations of Statistics

- Important exam points:
- Deals only with aggregates, not individuals
- Requires quantitative data
- Results may be misleading if data is wrong
- Cannot replace common sense
- Shows association, not causation



average
or
aggregate

Types of Data

- **(A) Primary Data**
- Collected first-hand by investigator.

for their own purpose.

- Methods:
- Observation
- Interview
- Questionnaire
- Schedule

(B) Secondary Data

- Already collected by others.
- Sources:
- RBI reports
- Government publications
- Annual reports
- Journals

etc.

Census vs Sampling

Census	Sampling
Entire population studied	Only part studied
More accurate	Less costly
Time consuming	Faster
Expensive	Economical

Classification of Data

- Arranging data into groups.
- Types:
- Qualitative
- Quantitative
- Chronological
- Geographical

Homogenous
groups

Time series

QQCG

Tabulation

- Systematic arrangement of data in rows and columns.
- Advantages:
- Easy understanding
- Comparison becomes simple
- Saves time
- Helps analysis

+

Assign Frequency

Frequency Distribution

- Shows number of observations in each class.
- Types:
- Discrete frequency distribution
- Continuous frequency distribution
- Important Terms:
- Class interval
- Frequency
- Mid value
- Cumulative frequency

Table f

By default

Class Mark

$\frac{0+5}{2} = 2.5$

	f	c.f
0-5	3	3
5-10	5	8
10-15	6	14
15-20	7	21

Important Definitions

- **Population**
- Entire group under study.
- **Sample**
- Part of population selected for study.
- **Parameter**
- Measure relating to population.
- **Statistic**
- Measure relating to sample.

1. A national research council in India launched a large analytical program to improve decision-making across multiple sectors including banking, healthcare, weather prediction, sports analytics and business strategy.

During the project, different departments were asked to apply statistical tools to their data sets.

- The banking division analysed customer deposits, loan demand, credit ratios and profitability patterns before modifying its credit policy.
- The healthcare division conducted clinical research comparing the effectiveness of multiple treatment combinations for a chronic disease.
- The meteorological unit attempted to forecast rainfall patterns using regression models and time series techniques.
- The business analytics team evaluated several investment strategies using Bayesian decision models to determine the optimal course of action.
- Meanwhile, a sports analytics unit examined historical player performance data to improve team strategy and correct recurring mistakes.

Before performing advanced analysis, the project leaders instructed all departments to properly organise their raw data sets into structured tables and categories. After completing mathematical computations, the research teams prepared policy recommendations for government and institutional decision-makers.

Based on the above situation, consider the following statements:

1. The step where raw datasets are organised into structured tables and groups corresponds to classification and tabulation in the statistical investigation process.
2. The meteorological unit's use of regression and time series techniques represents statistical analysis applied for predictive purposes.
3. The healthcare division's clinical trials demonstrate the application of statistics in evaluating competing treatment alternatives and identifying optimal combinations.
4. The banking division's modification of credit policies based on ratios and profitability analysis illustrates the use of probability-based statistical reasoning in financial decision-making.
5. The final preparation of policy recommendations by the research teams represents the interpretation stage of statistical investigation.
6. The sports analytics unit's study of historical player performance contradicts the fundamental objective of statistics because sports performance is qualitative rather than quantitative.
7. The use of Bayesian decision models in business strategy is intended to evaluate payoffs associated with alternative decisions.

Which of the following is correct?

A. 1, 2, 4, 5 and 6 only
B. 1, 2, 3, 4, 5 and 7 only
C. 2, 3, 4, 5, 6 and 7 only
D. 1, 2, 3, 4, 5, and 6 only

Answer: B

Solution:

Statement 1 – Correct: Classification and tabulation organise raw data into meaningful categories and tables, making the data suitable for statistical calculations.

Statement 2 – Correct: Regression analysis and time series techniques are commonly used for forecasting variables such as rainfall or weather conditions.

Statement 3 – Correct: Clinical research frequently relies on statistical techniques to evaluate treatment effectiveness and determine optimal therapeutic combinations.

Statement 4 – Correct: Banking decisions such as credit policy formulation often rely on statistical evaluation of ratios, profitability trends, and probability-based risk assessments.

Statement 5 – Correct: After analysis, conclusions and policy implications are drawn. This stage represents interpretation, where findings are translated into actionable insights.

Statement 6 – Incorrect: Sports performance can be quantified through statistics such as runs scored, goals, averages, strike rates, or win ratios. Statistical analysis is widely used in sports.

Statement 7 – Correct: Bayesian decision theory helps decision-makers evaluate the expected payoff of different alternatives under uncertainty and choose the optimal strategy.

2. A research team of a public policy think tank is analysing socio-economic conditions in a large metropolitan region. The researchers are interested in understanding patterns of income distribution, employment characteristics, and demographic indicators. Since collecting data from the entire population of the city is extremely expensive and time-consuming, the team decides to collect information from 5,000 households selected across different wards.

During the study, the following variables are recorded for each household:

- Number of earning members in the household
- Monthly household income (in ₹)
- Religion of the household head
- Height of the primary earning member (in cm)
- Number of accidents reported in the locality during the year

average, on sample → to estimate parameters for pop'

Using the sample data, the researchers compute the sample mean income and sample standard deviation to estimate the corresponding characteristics for the entire population of the city. However, a review committee raises certain concerns regarding the interpretation and limitations of the statistical results.

Consider the following statements regarding the concepts and issues involved in the study:

1. The entire set of households in the metropolitan region constitutes the population for the study, whereas the 5,000 selected households form the sample.
2. The number of earning members and number of accidents are examples of discrete variables, while monthly income and height are examples of continuous variables.
3. Religion of the household head is treated as an attribute because it represents a qualitative characteristic that cannot be expressed numerically.
4. The sample mean income calculated from the 5,000 households represents a parameter because it summarizes the population characteristic.
5. Statistical analysis based on the collected data may provide approximate conclusions about the population but cannot guarantee exact results for every individual unit.
6. If the data collection process is influenced by enumerators who deliberately exclude certain low-income households, the statistical results may become biased.
7. Statistical methods can meaningfully analyze the income of a single household without requiring any comparison with other observations.

Which of the following options is correct?

A. 1, 2, 3, 5 and 6 only

B. 1, 2, 3, 4 and 6 only

C. 1, 2, 3, 5, 6 and 7 only

D. 1, 3, 4, 5 and 7 only

N-1

Answer: A

Solution:

Statement 1 – Correct: Population refers to the entire group of interest. Here, all households in the metropolitan region form the population, while the 5,000 households selected for observation constitute the sample.

Statement 2 – Correct: Discrete variables take countable values.

Number of earning members and number of accidents are countable: Discrete variables. Continuous variables can take any value within a range, including fractions. Monthly income and height can vary across a continuous range : Continuous variables.

Statement 3 – Correct: Religion cannot be measured numerically. Such qualitative characteristics are called attributes.

Statement 4 – Incorrect: The sample mean is not a parameter. A parameter describes a population (e.g., population mean). A statistic describes a sample (e.g., sample mean). Thus the sample mean income is a statistic, used to estimate the population parameter.

Statement 5 – Correct: Statistical results derived from samples provide approximate results on average, especially when only a subset of the population is analysed.

Statement 6 – Correct: If enumerators exclude certain groups (e.g., low-income households), the data collection becomes biased, leading to misleading conclusions.

Statement 7 – Incorrect: Statistics deals with aggregates or mass data, not individual observations. A single household's income cannot be statistically analysed without comparison with other observations.

3. A research institute is conducting a nationwide study on financial literacy among households. The project team adopts multiple approaches to gather information.

- Initially, investigators conduct face-to-face interviews with selected households to understand their financial behaviour.
- They also design an online questionnaire containing both multiple-choice and descriptive questions which is circulated to thousands of respondents.
- Due to budget constraints, the researchers collect detailed responses only from 5,000 households out of the total 2 million households, ensuring that age distribution, gender composition, and urban-rural proportion match the national structure.
- For historical comparison, the researchers use data published earlier by RBI, WHO reports on financial awareness indicators, and a past government census report.
- Later, another academic researcher uses the final report of this study as a source for his own research.

Based on the above situation, evaluate the following statements:

1. The face-to-face interaction conducted with households constitutes the Direct Interview Method of collecting primary data.
2. The online questionnaire used in the study cannot be treated as primary data because it is administered through the internet.
3. The selection of 5,000 households out of 2 million households represents a sample survey, provided the sample reflects population characteristics.
4. The earlier reports of RBI and WHO used in the study constitute secondary data for the research institute.
5. If the entire 2 million households had been surveyed instead of 5,000 households, the exercise would have been classified as a census.
6. The final report prepared by the institute will continue to remain primary data for all future researchers who use it.
7. When another academic researcher uses the institute's report for his study, the information becomes secondary data for that researcher.

Which of the above statements are correct?

A. 2, 3, 4, 6 and 7 only
B. 1, 2, 3, 4 and 5 only
C. 1, 3, 5, 6 and 7 only
D. 1, 3, 4, 5 and 7 only

S - 12

Answer: D

Solution:

Statement 1 – Correct. Conducting face-to-face interaction with respondents for information collection is the Direct Interview Method, a primary data collection technique.
Statement 2 – Incorrect. The mode of administration (online or offline) does not determine whether data is primary or secondary. If researchers collect responses directly through questionnaires, it remains primary data.
Statement 3 – Correct. Collecting information from 5,000 households out of 2 million is a sample survey, especially when the sample mirrors the population's demographic structure.
Statement 4 – Correct. Data taken from existing reports of organizations like RBI or WHO were previously collected for other purposes, making them secondary data for the present research.
Statement 5 – Correct. Surveying every household in the population would constitute a census, as data would be collected from all units rather than a subset.
Statement 6 – Incorrect. The report prepared by the institute is primary only for the institute that collected the data. For others who later use it, the same information becomes secondary data.
Statement 7 – Correct. When another researcher uses the institute's report, the data are second-hand for him, hence secondary data.

4. A research analyst is examining the demographic and performance characteristics of employees in a large financial institution. The raw dataset contains 2,400 employee records including age, monthly income, gender, city of posting, year of joining, and training score obtained in an internal certification test.

To analyse the dataset effectively, the analyst performs several steps. First, employees are grouped by gender and literacy level of their highest qualification. In another exercise, the analyst arranges employees according to their city of posting across different regions of the country. In a separate table, the analyst arranges the number of employees joining the bank in each year from 2015 to 2024.

For analysing the training scores, the analyst constructs the following frequency distribution:

Training Score	Frequency
40–49	6
50–59	10
60–69	18
70–79	26
80–89	20
90–99	12

$-1 \div 2 = \boxed{0.5}$

(−) 0.5

(+) 0.5

$16 + 18 = 34$

$21 + 10$

5.26

Later, the analyst converts the class interval 60–69 into an exclusive class boundary form, and also calculates relative frequency, percentage frequency, and frequency density for some classes.Based on the above information, consider the following statements:

1. Classification of employees according to gender and literacy level represents a two-way classification.
2. Arranging employees according to city of posting across regions represents geographical classification.
3. The table showing number of employees joining the bank from 2015–2024 represents chronological classification.
4. The class interval 60–69 is an inclusive type class interval, and its exclusive class boundary form will be 59.5–69.5.
5. The relative frequency of the class interval 70–79 is approximately 0.283.
6. The percentage frequency of the class interval 80–89 is approximately 21.74%.
7. The frequency density of the class interval 50–59 is 1.5.

$RF = \frac{CF}{TF} = \frac{26}{92}$

Which of the following is correct?

$\frac{C.F}{width} = \frac{10}{9}$

$\% = \%\,Relative = \frac{20}{92} \times 100$

A. 1, 2, 3, 5 and 6 only
B. 1, 2, 4, 5 and 6 only
C. 1, 2, 3, 4, 5 and 6 only
D. 1, 2, 3, 4, 5, 6 and 7

Answer: C

Solution:

Statement 1 – Correct: Two-way classification involves two characteristics simultaneously. Example here: Gender, Literacy level of qualification. Thus classification based on these two attributes together forms two-way classification.

Statement 2 – Correct: When data is organised according to geographical regions such as cities, states or countries, it represents geographical classification. Grouping employees according to city of posting across regions satisfies this criterion.

Statement 3 – Correct: Chronological classification arranges data according to time order such as: years, months, weeks. The table showing employees joining from 2015–2024 is therefore chronological (time series) classification.

Statement 4 – Correct: The interval 60–69 is an inclusive interval because both limits belong to the class. Conversion to exclusive class boundary: Gap between classes = 70 – 69 = 1; Half correction factor = 1 / 2 = 0.5. New limits: Lower boundary = 60 – 0.5 = 59.5; Upper boundary = 69 + 0.5 = 69.5. So class becomes: 59.5 – 69.5.

Statement 5 – Correct: Relative frequency formula:

$$Relative\ Frequency = \frac{Class\ Frequency}{Total\ Frequency}$$

For 70–79 class: Frequency = 26. Total = 92. 26/92 = 0.2826≈ 0.283

Statement 6 – Correct: Percentage frequency:

$$\frac{Frequency}{Total} \times 100$$

For 80–89 class: 20/92×100=21.739≈21.74%

Statement 7 – Incorrect: Frequency density formula:

$$Frequency\ Density = \frac{Class\ Frequency}{Class\ Width}$$

For 50–59 class. Frequency = 10

Class width: 59−50=9

However, in grouped data inclusive classes converted to exclusive boundaries give width: 49.5–59.5 = Width = 10

So,10/10=1

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UNIT 2 Sampling Techniques

STRUCTURE

Average $\bar{x}$

Sample A B C D E

Strata

Homogenous group

$n \geq 30$

Sampling

N = Popul

< 5

UNIT 2 – Sampling Techniques

- **1. Meaning of Sampling**
- Sampling means selecting a representative portion from the population.
- Used because:
- Census is costly
- Census consumes time
- Large populations are difficult to study fully

Advantages of Sampling

- Economical
- Faster
- Practical
- Requires less manpower
- Useful in banking surveys

Types of Sampling

- **A. Probability Sampling**
- Each unit has equal chance of selection.
- **Types:**
- **(i) Simple Random Sampling**
- Selection through lottery/random numbers.
- **(ii) Systematic Sampling**
- Every nth item selected.
- Example:
  Every 10th customer.

- **(iii) Stratified Sampling**
- Population divided into homogeneous groups.
- Example:
  Borrowers divided into:
  - Agriculture
  - MSME
  - Retail
- **(iv) Cluster Sampling**
- Population divided into clusters.

B. Non-Probability Sampling

- **(i) Convenience Sampling**
- Easy-to-reach units selected.
- **(ii) Judgment Sampling**
- Selection based on investigator judgment.
- **(iii) Quota Sampling**
- Fixed quotas assigned.

Sampling Errors

- Difference between sample result and population result.
- Types:
- Random error
- Systematic error

$\bar{x} = 90$ $\bar{X} = 100$

Standard Error (SE)

V.V.V imp

↑ sample ↓ SE

↑ sample → Better Results.

- Very important CAIIB concept.
- Standard Error measures:
  Variation between sample statistic and population parameter.
- Lower SE = Better sample reliabilit

$$S.E = \frac{\sigma_{population}}{\sqrt{n}}$$

N – pop

n = sample.

Central Limit Theorem (CLT)

- If sample size is sufficiently large:
- Sampling distribution becomes normal
- Mean of sample means equals population mean
- Very important numerical topic.

Characteristics of Good Sample

- Representative
- Adequate size
- Free from bias
- Scientifically selected

Exam-Oriented Formulas

- **Standard Error of Mean**
- $\boxed{SE = \frac{\sigma}{\sqrt{n}}}$
- Where:
- σ = Standard deviation
- n = Sample size

1. A multinational dairy company processes 1,20,000 litres of milk per day across several processing units. Testing every milk packet for quality is impractical. The quality control department selects random packets from each batch and tests them for bacterial contamination and fat content.

At the same time, the marketing division is conducting a survey to understand consumer migration preferences for dairy brands in urban India. Initially, they ask only large supermarket chains and premium consumers for their opinions. Later, a statistician advises that the survey should be redesigned to use a random probability sample covering households across different income groups.

The company also wants to estimate the average fat content of all milk packets produced during a day using the results obtained from the sampled packets.

Based on the above situation, consider the following statements:

1. The entire day's production of milk packets constitutes the population for the quality testing study.
2. The packets selected for laboratory testing represent the sampling units, and the number of packets selected forms the sample size.
3. The average fat content calculated from the tested packets is a population parameter because it reflects the entire production output.
4. The initial marketing survey that targeted only supermarket chains and premium consumers represents a biased sample.
5. Random probability sampling would allow the company to statistically estimate the reliability of results, unlike judgement sampling.
6. Judgement sampling cannot be used even for preliminary pilot studies in market research.
7. If the company conducted quality tests on every single packet produced during the day, the process would be called complete enumeration or census.

Which of the following is correct?

A. 1, 2, 4, 5 and 7 only
B. 1, 3, 4, 5 and 6 only
C. 2, 3, 4, 6 and 7 only
D. 1, 2, 3, 5 and 7 only

Answer: A

Solution:

Statement 1 – Correct: The population refers to all items under investigation. In this case, the entire day's milk production represents the population.

Statement 2 – Correct: Each tested milk packet is a sampling unit, and the total number selected constitutes the sample size.

Statement 3 – Incorrect: The mean fat content derived from sampled packets is a statistic, not a parameter. A parameter would describe the entire population.

Statement 4 – Correct: Surveying only premium consumers and supermarket chains creates selection bias, as the sample does not represent the broader population of dairy consumers.

Statement 5 – Correct: Random probability sampling enables statistical inference, allowing reliability and sampling error to be measured.

Statement 6 – Incorrect: Judgement sampling can be used for pilot or preliminary studies before designing a random sample.

Statement 7 – Correct: Testing every packet would mean data from the entire population, which is known as complete enumeration or census.

2. A research analyst in a consulting firm wants to study employee work–life balance. The firm has 100 employees numbered from 00 to 99. To select a simple random sample of 10 employees, the analyst decides to use a table of random digits.

Procedure adopted:

- Each employee is assigned a two-digit number from 00 to 99.
- The analyst reads the random digit table row-wise and selects the first two digits of each five-digit number (e.g., 15, 20, 82, 83 …).
- If a number already selected appears again, it is ignored and the next number is considered, since the sampling is without replacement.

Evaluate the following statements:

1. In simple random sampling, every possible sample of size 10 from the population of 100 employees has an equal probability of being selected.
2. If the analyst had adopted sampling with replacement, the same employee could appear more than once in the final sample.
3. If the first ten two-digit numbers obtained from the random digit table are 15, 20, 82, 83, 69, 09, 72, 85, 48, 06, all of them can directly constitute the sample without any rejection under the stated procedure.
4. In simple random sampling without replacement from a population of 100 employees, the probability that a specific employee is included in a sample of 10 employees equals 0.10.
5. The use of a table of random digits ensures that each digit from 0–9 appears with equal probability, and every sequence of equal length is equally likely.
6. If the sampling process were conducted with replacement, the probability that a particular employee appears at least once in a sample of 10 draws would still remain exactly 0.10.

Which of the following is correct?

A. 1, 2, 3, 4 and 5 only
B. 1, 2, 4 and 5 only
C. 1, 2, 3, 4, 5 and 6
D. 1, 3, 4 and 6 only

Answer: B

Solution:

Statement 1 — Correct: Simple Random Sampling (SRS) ensures that every possible sample of a fixed size has equal probability of selection. Thus, every combination of 10 employees from the 100 employees has the same chance of being selected.

Statement 2 — Correct: Under sampling with replacement, the selected unit is placed back into the population before the next draw. Hence the same employee can be selected more than once.

Statement 3 — Incorrect: Although all numbers (15, 20, 82, 83, 69, 09, 72, 85, 48, 06) lie within the valid range 00–99, the sampling procedure requires checking for duplicate numbers because the sampling is without replacement. If a repeated number appears, it must be ignored and another number selected. Therefore, the first ten numbers cannot automatically form the final sample without verifying duplicates.

Statement 4 — Correct: For Simple Random Sampling Without Replacement (SRSWOR): P(a particular employee included)=n/N

Where: n=10 (sample size); N=100 (population); P=100/10=0.10

Statement 5 — Correct: Random digit tables are constructed so that:

- Each digit 0–9 appears with equal probability.
- Any sequence of digits of the same length has equal likelihood of occurrence.

This ensures true statistical randomness in sampling.

Statement 6 — Incorrect Under sampling with replacement, the probability that a particular employee appears at least once in 10 draws is: P(at least once)=1−P(never selected)

Probability of not selecting that employee in one draw: 99/100

Probability of never selecting in 10 draws: $(99/100)^{10}$

Thus: P(at least once)=$1-(99/100)^{10} \approx 0.0956$. This is not equal to 0.10, so the statement is incorrect.

3. A nationwide market research agency is studying consumer preferences for smart televisions across a metropolitan region with 5 million households.

The research director evaluates different sampling methods:

- The city is divided into 1,000 residential blocks, each containing approximately 500 households. The team randomly selects 40 blocks and surveys all households in those blocks.
- Another research analyst proposes dividing households into income groups (low, middle, upper) and drawing samples proportionate to their population share.
- A junior analyst suggests selecting every 25th household from the municipal registry after choosing a random starting point.
- The research director also notes that many statistical inference procedures assume the sample originates from simple random sampling even if practical sampling differs.
- During pilot testing, the analyst observes that selecting households every 7th entry in a street register leads to over-representation of apartment complexes due to the ordering pattern.

Based on the above scenario, evaluate the following statements:

1. The method of selecting 40 residential blocks and surveying every household within them represents Cluster Sampling.
2. Dividing households into income groups and sampling proportionately represents Stratified Sampling designed to reduce sampling error when strata are internally homogeneous.
3. Selecting every 25th household after choosing a random start represents Systematic Sampling.
4. If the ordering of the municipal registry follows a hidden pattern linked to housing type, systematic sampling may produce biased results.
5. The cluster sampling design assumes that each cluster is broadly representative of the population characteristics.
6. Stratified sampling is most efficient when there is large variation within each stratum but minimal variation between strata.
7. Statistical inference procedures commonly assume Simple Random Sampling, even when other sampling methods are operationally used.

Which of the following is correct?

A. 2, 3, 4, 5, 6 and 7 only

B. 1, 2, 3, 5 and 6 only

C. 1, 2, 3, 4, 5 and 7 only

D. 1, 3, 4, 5 and 6 only

Answer: C

Solution:

Statement 1 – Correct: Selecting clusters (blocks) and surveying all households inside them corresponds to cluster sampling.

Statement 2 – Correct: Stratified sampling divides the population into homogeneous strata (income groups) and samples proportionally, improving representativeness.

Statement 3 – Correct: Choosing every 25th household after a random start fits the definition of systematic sampling.

Statement 4 – Correct: Systematic sampling can become biased when the population list contains a periodic structure that aligns with the sampling interval.

Statement 5 – Correct: Cluster sampling assumes each cluster resembles the population as a whole.

Statement 6 – Incorrect: The described condition actually corresponds to cluster sampling. Stratified sampling is most efficient when variation within strata is small and variation between strata is large.

Statement 7 – Correct: Many theoretical statistical inference techniques assume simple random sampling as the base framework.

4. A risk analytics team in a large bank is studying the age of non-performing assets (NPAs) in years. The population of NPAs has an unknown distribution with mean μ and standard deviation σ.

Instead of examining the entire population, the analysts repeatedly draw random samples of 10 NPAs each from the portfolio. For every sample, they calculate the sample mean age of NPAs.

After conceptualizing all possible samples of size 10, they visualize the distribution of these sample means. This theoretical distribution has mean μ_x and standard deviation S_x.

$\frac{x+y+z}{3}$

Based on the above scenario, evaluate the following statements:

1. The distribution formed by all possible sample means of size 10 is known as the sampling distribution of the mean.
2. The mean of this sampling distribution μ_x is expected to be approximately equal to the population mean μ.
3. The standard deviation S_x of the sampling distribution represents the standard error of the mean.
4. The variability in the sample means arises primarily because different random samples contain different combinations of NPAs.
5. If the analysts increase the sample size from 10 to 50, the standard error of the mean would generally decrease.

Which of the following options is correct?

A. 1, 2, 3 and 4 only
B. 1, 2, 3, 4 and 5
C. 1, 2, 3 and 5 only
D. 1, 3, 4 and 5 only

only if normal distribution

$\frac{\sigma}{\sqrt{n}}$

Answer: B

Solution:

Statement 1 – Correct: The sampling distribution of the mean is constructed from the means of all possible samples of a fixed size drawn from a population.

Statement 2 – Correct: For unbiased random sampling, the expected value of the sample mean equals the population mean. Hence the center of the sampling distribution coincides with the population mean.

Statement 3 – Correct: The standard deviation of the sampling distribution of the sample mean is defined as the standard error of the mean.

Statement 4 – Correct: Different samples contain different elements from the population. This creates random variability in the computed sample means.

Statement 5 – Correct: Standard error generally equals: $$SE = \frac{\sigma}{\sqrt{n}}$$

When the sample size increases, the denominator becomes larger, causing the standard error to decline. Hence larger samples produce more stable estimates.

5. A large commercial bank analyses the distribution of balances in its savings accounts. The balances follow a normal distribution with:

Mean balance = ₹2,000

Standard deviation = ₹600

The bank's analytics department randomly selects samples of 100 accounts to study the behaviour of average balances. The analysts want to evaluate the probability that the sample mean balance lies between ₹1,900 and ₹2,050. Using the standard normal transformation the calculated Z-values are:

For ₹1,900: Z = −1.67 → 0 to 1.67

For ₹2,050: Z = 0.83 → .7242

(+)

Z = .4525

$\frac{SD}{\sqrt{n}} = \frac{600}{\sqrt{100}} = \frac{600}{10} = 60$

Based on this information, evaluate the following statements:

1. The standard error of the mean for the sample of 100 accounts equals ₹6.
2. The sampling distribution of the mean will follow a normal distribution with mean ₹2,000 and standard deviation ₹60.
3. The probability that the sample mean lies between ₹1,900 and ₹2,050 equals 0.8749.
4. If the sample size were increased to 400, the standard error would fall to ₹30.
5. If the population standard deviation doubled while sample size remained unchanged, the standard error would also double.
6. The probability between Z = −1.67 and Z = 0.83 is obtained by adding the areas because the Z values have opposite signs.
7. If both Z-values were positive (for example 0.83 and 1.67), the probability between them would be found by subtracting the smaller table value from the larger one.

Error

Which of the following is correct?

$n_1 = 100 \rightarrow SE = 60$

$n_2 = 400 \rightarrow SE = 30$

$\frac{600}{\sqrt{400}} = \frac{600}{20} = 30$

A. 1, 2, 3, 4, 5, 6 and 7

B. 1, 2, 3, 4, 6 and 7 only

C. 1, 2, 4, 5 and 6 only

D. 2, 4, 5, 6 and 7 only

Answer: D

Solution:

Statement 1 – Incorrect: $SE = \frac{\sigma}{\sqrt{n}}$ = 600/10 = 60

Statement 2 – Correct: Sampling distribution parameters: Mean = 2000 Standard deviation (SE) = 60. Thus: $\bar{X} \sim N(2000, 60)$

Statement 3 – Incorrect: Probability between −1.67 and 0.83: Area (0 to 1.67) = 0.4525 Area (0 to 0.83) = 0.2967

Because signs are opposite: 0.4525 + 0.2967 = 0.7492. 0.4525+0.2967=0.7492 Thus probability = 0.7492 (74.92%)

Statement 4 – Correct: If n = 400, 600 / 20=30. Standard error reduces as sample size increases.

Statement 5 – Correct: If σ doubles: $SE = \frac{\sigma}{\sqrt{n}}$. Hence SE also doubles.

Statement 6 – Correct: For opposite signs, areas are added.

Statement 7 – Correct If both Z values are positive: P(0.83<Z<1.67). Then: Area(1.67)−Area(0.83)

Thus subtraction rule applies.

6. A statistics researcher is analysing the life of motorcycle tyres (in months) for five owners: Chetan (3), Dinesh (3), Eswar (6), Feroz (9) and George (15). The population distribution is clearly non-normal and highly skewed because of the extreme value (15 months).

To understand the behaviour of the sampling distribution of the mean, the researcher considers all possible samples of size 3 without replacement from the population and computes the sample means. After calculating all possible combinations, the researcher observes that:

- Total population tyre life = 36 months
- Population mean = 7.2 months
- Sum of all calculated sample means = 72
- Number of samples = 10

$S_1 + S_2 + S_3 + S_{10}$ $\quad \frac{72}{10} = 7.2 \quad \frac{36}{5} = 7.2$

The researcher also observes that the distribution of the sample means appears more symmetric than the original population distribution.

Based on the above situation, consider the following statements:

1. Even though the population distribution is non-normal, the mean of the sampling distribution of the sample mean will still equal the population mean if all possible samples are considered.
2. The sampling distribution of the mean will always have exactly the same shape as the population distribution, regardless of sample size.
3. As the sample size increases, the sampling distribution of the sample mean tends to approach a normal distribution even if the population distribution is skewed.
4. In the above case, the mean of the sampling distribution equals 7.2 months because the sum of all sample means divided by the number of samples equals the population mean.
5. If the sample size were increased significantly and many samples were drawn, the Central Limit Theorem would explain why the sampling distribution becomes approximately normal.

Which of the following options is correct?

A. Only 1, 3 and 4 are correct
B. Only 1, 3, 4 and 5 are correct
C. Only 2, 3 and 5 are correct
D. Only 1, 2, 3 and 5 are correct

Answer: B

Solution:

Statement 1: Correct: The expected value of the sample mean equals the population mean, regardless of the population's distribution. $E(\bar{X}) = \mu$

Thus, even when the population is skewed or non-normal, the mean of the sampling distribution equals the population mean when all samples are considered.

Statement 2: Incorrect: The sampling distribution does NOT always have the same shape as the population distribution.

In fact: For small samples, it may resemble the population. As sample size increases, it becomes more normal. Thus the statement contradicts statistical theory.

Statement 3: Correct. This reflects the Central Limit Theorem (CLT) principle. When sample size increases:

- Sampling distribution of mean: approximately normal
- This holds irrespective of population shape

Hence, even skewed populations eventually produce near-normal sampling distributions.

Statement 4: Correct. Here: Sum of sample means=72; Number of samples=10 $\mu_{\bar{X}}$=72/10=7.2. Which equals the population mean: μ=36/ 5=7.2. Thus the sampling distribution mean equals the population mean.

Statement 5: Correct: This statement reflects the Central Limit Theorem, which states: For large samples, the sampling distribution of the mean approaches normality, regardless of population distribution.

- Therefore, it correctly explains the phenomenon observed in the case.

7. A large private bank is analysing the monthly incentive earnings of relationship managers across the country. The population distribution of incentive earnings is highly positively skewed because a few managers earn extremely high incentives. Historical data show that:

Population mean incentive = ₹52,000

Population standard deviation = ₹9,000

The HR analytics team randomly selects a sample of 36 managers to estimate the average incentive payout. n

Using the Central Limit Theorem, they evaluate the probability that the sample mean incentive exceeds ₹55,000.

The team computes the standard error and converts the result into a z-score to estimate the probability from the standard normal distribution.

Consider the following statements:

1. Even though the population distribution of incentives is positively skewed, the sampling distribution of the sample mean will tend to approximate a normal distribution because the sample size is sufficiently large.
2. The standard error of the sample mean for the given sample size equals ₹1,500. $\rightarrow \frac{\sigma}{\sqrt{n}} = \frac{9000}{\sqrt{36}} = 1500$
3. The calculated z-value for the sample mean of ₹55,000 is approximately 2.00.
4. The probability that the sample mean exceeds ₹55,000 is more than 3%.
5. If the sample size were increased from 36 to 144, the standard error would reduce to one-half of its previous value.

Which of the above statements are correct?

A. 1, 2 and 3 only

B. 1, 3 and 5 only

C. 1, 2, 3 and 5 only

D. 1, 2, 4 and 5 only

$$\frac{9000}{\sqrt{144}} = \frac{9000}{12} = 750$$

$$\boxed{Z = \frac{\bar{x} - \mu}{S.E}} = \frac{55000 - 52000}{1500} = \frac{3000}{1500} = 2$$

Answer: C

Solution:

Statement 1: Correct: The Central Limit Theorem (CLT) states that the sampling distribution of the sample mean approaches normality as the sample size increases, regardless of the shape of the population distribution. Since n = 36 (>30), the sampling distribution of the mean can be treated as approximately normal, even though the population distribution is positively skewed.

Statement 2: Correct: Standard Error of Mean: $SE = \frac{\sigma}{\sqrt{n}}$ = 9000 / 6 = 1500

Statement 3: Correct: $z = \frac{\bar{x} - \mu}{SE}$ = 3000 / 1500 = 2

Statement 4: Incorrect: For z = 2. Area between mean and z ≈ 0.4772. Right-tail probability: 0.5 − 0.4772 = 0.0228. Probability ≈ 2.28%

Statement 5: Correct: New sample size: n=144. SE = 9000 / 12 = 750. Thus standard error becomes exactly half, showing the inverse square-root relationship between sample size and standard error.

8. A policy research unit is analysing employee attrition across a finite population of specialised manufacturing firms. The population consists of N = 120 firms operating in a niche industry. Historical data shows that the standard deviation of annual employee turnover across the population is 84 employees.

The research team randomly selects n = 30 firms without replacement to estimate the mean turnover level for policy recommendations. A senior statistician suggests adjusting the standard error of the sample mean using the finite population multiplier (FPM) because the sampling fraction is not negligible.

Using the relevant statistical relationships, evaluate the following statements:

1. The sampling fraction for the study is 0.25, which is large enough to justify the use of the finite population multiplier.
2. The finite population multiplier for this sample is approximately 0.868.
3. The standard error of the sample mean without the finite population adjustment would be approximately $\boxed{\text{15.34 employees.}}$
4. After applying the finite population multiplier, the adjusted standard error of the mean becomes approximately 13.31 employees.
5. If the researchers instead sampled 9 firms, the finite population multiplier could safely be ignored under the generally accepted rule in sampling theory.

Which of the following is correct?

A. 1, 2, 3 and 4 only
B. 1, 3 and 4 only
C. 1, 2, 3, 4 and 5
D. 2, 3 and 4 only

$\sqrt{\frac{N-n}{N-1}}$ $\sqrt{\frac{120-30}{120-1}}$ 25% FPM $\frac{\sigma}{\sqrt{n}}$ $\frac{84}{\sqrt{30}}$ 25 > 5 FPM

S.E × $\sqrt{\frac{N-n}{N-1}}$ = 15.34 × .868

Answer: A

Solution:

Statement 1: Correct: Sampling fraction n/N=30/120=0.25. Sampling fraction = 25% of the population

Statistical rule:

- If sampling fraction < 0.05, FPM can be ignored.
- If ≥ 0.05, FPM should be applied.

Since 0.25 > 0.05, the adjustment is required.

Statement 2: Correct: Finite Population Multiplier formula: $FPM = \sqrt{\frac{N-n}{N-1}}$

Hence the multiplier is about 0.868–0.869.

Statement 3: Correct: Standard error assuming infinite population: $SE = \frac{s}{\sqrt{n}}$ = SE≈15.34. So the unadjusted standard error ≈ 15.34 employees.

Statement 4: Correct

Adjusted standard error using FPM: $SE_{adjusted} = \frac{s}{\sqrt{n}} \times FPM$ =15.34×0.868 ≈13.31. Thus adjusted SE ≈ 13.31 employees.

The multiplier reduces the sampling error because sampling occurs without replacement from a finite population.

Statement 5: Incorrect: If n = 9. Sampling fraction:

9/120=0.06

The rule states that the multiplier may be ignored when sampling fraction is less than 0.05.

Hence the statement is incorrect.

JAIIB CAIIB
PREP

UNIT 3 Measures of Central Tendency & Dispersion, Skewness, Kurtosis

STRUCTURE

10 Min Break

Ready at 6:30

Unit 3

PamB

Part of CT

1) Average — AM, GM, HM

2) Middle → Median
odd — even

3) Highest freq. → Mode

Disp.
eg. S.D, MD, Range

Skewness

Kurtosis

UNIT 3 – Measures of Central Tendency & Dispersion, Skewness, Kurtosis

- **PART A – Measures of Central Tendency**
- Central tendency means a single value representing entire data.

- **1. Arithmetic Mean (AM)**
- Most commonly used average.
- Formula:
- $\bar{x} = \frac{\sum x}{n}$
- Advantages:
- Simple
- Uses all observations
- Disadvantages:
- Affected by extreme values

Geometric Mean (GM)

- **Definition**
- Geometric Mean is the average obtained by multiplying all observations and then taking the nth root.
- Used when:
- growth rates are involved,
- ratios or percentages are involved,
- compound increase/decrease calculations.
- Examples:
- Population growth
- Investment returns
- Banking growth rates

Formula of Geometric Mean

- $GM = \sqrt[n]{x_1 x_2 x_3 \cdots x_n}$
- Where:
- n= number of observations
- $x_1, x_2, \ldots$ =observations

- **Example of GM**
- Find GM of:
  2, 8
- Solution:
- $GM = \sqrt{2 \times 8} = \sqrt{16} = 4$

Features

Point	Explanation
Uses all observations	Yes
Affected by extreme values	Less than AM
Suitable for growth rates	Yes
Cannot be used if value is zero/negative	Yes

Harmonic Mean (HM)

- **Definition**
- Harmonic Mean is the reciprocal of the arithmetic mean of reciprocals.
- Used when:
- average speed is required,
- rates and ratios are involved.
- Examples:
- Speed calculations
- Average price per unit
- Financial ratios

Formula of Harmonic Mean

- $HM = \frac{n}{\sum \frac{1}{x}}$
- Where:
- n= number of observations
- x= observations

Example of HM

- Find HM of:
  2 and 4
- Solution:
- $HM = \frac{2}{\frac{1}{2}+\frac{1}{4}} = \frac{2}{\frac{3}{4}} = \frac{8}{3}$
- So:
  HM = 2.67

Median

- Middle value after arranging data.
- Useful when:
- Extreme values exist
- Income distribution analysis

Mode

- Most frequently occurring value.
- Used in:
- Market research
- Consumer preference

Geometric Mean (GM)

- Used for:
- Growth rates
- Compound returns
- Formula:
- $GM = \sqrt[n]{x_1 x_2 x_3 \ldots x_n}$

Harmonic Mean (HM)

- Used for:
- Average speed
- Ratios
- Formula:
- $HM = \frac{n}{\sum \frac{1}{x}}$

Relationship Among Mean, Median & Mode

- Important exam formula:
- $Mode = 3Median - 2Mean$

PART B – Measures of Dispersion
Dispersion means spread of data.

- **1. Range**
- Difference between highest and lowest value.
- Formula:
- $Range = L - S$
- Where:
- L = Largest value
- S = Smallest value

- **2. Quartile Deviation**
- Based on quartiles.
- Formula:
- $QD = \frac{Q_3 - Q_1}{2}$

- **3. Standard Deviation (SD)**
- Most important dispersion measure.
- Measures consistency.
- Lower SD = More consistency.
- Formula:
- $\sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$

- **4. Coefficient of Variation (CV)**
- Used to compare consistency.
- Formula:
- $CV = \frac{\sigma}{\bar{x}} \times 100$
- Lower CV = Better consistency.

PART C – Skewness

Skewness measures asymmetry of distribution.

- **Types of Skewness**
- **Positive Skewness**
- Tail towards right
- Mean > Median > Mode
- **Negative Skewness**
- Tail towards left
- Mode > Median > Mean
- **Symmetrical Distribution**
- Mean = Median = Mode

- **PART D – Kurtosis**
- Measures peakedness of distribution.
- Types:
- Leptokurtic → Highly peaked
- Mesokurtic → Normal
- Platykurtic → Flat

1. A financial analyst is studying the monthly incentive earnings of employees in three departments of a company. The analyst initially calculated the mean incentive of Department A (40 employees) as ₹52 and Department B (60 employees) as ₹64.

Later, the following issues were identified:

1. In Department A, one incentive amount of ₹96 had been wrongly recorded as ₹69 while calculating the mean.
2. In Department B, the analyst realised that the incentive of one employee worth ₹120 had been omitted entirely from the dataset.
3. A third department (Department C) with 50 employees has a correctly computed mean incentive of ₹70.

After correcting the above errors and including the omitted value, the analyst decides to compute the corrected combined mean incentive for all three departments together.

Consider the following statements:

1. The original total incentive calculated for Department A before correction was ₹2,080.
2. After correcting the recording error, the revised mean incentive of Department A becomes ₹52.675.
3. The corrected mean incentive of Department B after including the omitted value becomes ₹64.92 (approx.).
4. The combined mean incentive of Departments A and B after corrections is approximately ₹59.02.
5. The final combined mean incentive of all three departments together after corrections is approximately ₹62.56.

Which of the above statements are correct?

A. 1, 2 and 3 only
B. 1, 2, 3 and 4 only
C. 2, 3 and 5 only
D. 1, 2, 3, 4 and 5

$\frac{Total}{40} = 52 \Rightarrow Total = 52 \times 40$

2080 incorrect
(−) 69
(+) 96
2107 ÷ 40

$40 \times 52.675 + 61 \times 64.92$

$\frac{N_1\bar{x}_1 + N_2\bar{x}_2}{N_1+N_2}$ 40+61

$\frac{Total}{60} = 64$

Total = 64 × 60 = 3840
(+) 120
3960

$\frac{3960}{61}$

Answer: A

Solution:

Step 1: Department A – Original Total Incentive

Mean = 52; Number of employees = 40; Total= $n \times \bar{X}$ = 40×52=2080

Step 2: Correcting the Misread Value

Wrong value = 69; Correct value = 96. Corrected total: 2080−69+96=2107. Corrected mean: 52.675

Step 3: Department B – Adding Omitted Observation

Original mean = 64; Employees counted = 60

Original total: 60×64=3840; Omitted value = 120; Corrected total: 3840+120=3960; Corrected number of employees:61 ; Corrected mean: 3960 / 61 = 64.92

Step 4: Combined Mean of Departments A and B

Corrected totals: Department A = 2107; Department B = 3960; Combined total: 2107+3960=6067; Total employees: 40+61=101. Combined mean: 6067/ 101 = 60.07

Step 5: Final Combined Mean of A, B, and C

Department C: Employees = 50; Mean = 70

Total: 50×70=3500; Grand total: 2107+3960+3500=9567

Total employees: 40+61+50=151. Final combined mean: 9567/ 151 = 63.357

2. A financial analyst is evaluating the long-term performance of four start-ups included in a venture portfolio. The annual growth multipliers of the companies over four years were recorded as 1.10, 1.24, 0.95 and 1.32.

Another intern mistakenly computed the Arithmetic Mean (AM) of the multipliers to estimate the average annual growth, while the senior analyst insisted that Geometric Mean (GM) should be used because the growth rates are compounded over time.

Assume that all values are positive and represent multiplicative growth factors.

Consider the following statements:

1. The Geometric Mean of the four multipliers will be less than the Arithmetic Mean of the same observations.
2. If any one of the growth multipliers were zero, the Geometric Mean of the entire series would also become zero.
3. If one of the multipliers were negative, the Geometric Mean would still exist but would require taking an even root of a negative product.
4. The Geometric Mean represents the constant annual compounded growth factor which, when applied for four years, produces the same overall growth as the actual multipliers.
5. If the analyst converts the multipliers to percentage growth rates and averages them using the Arithmetic Mean, the result will give the same compounded portfolio growth as the Geometric Mean.

Which of the following is correct?

A. 1, 2, 3 and 4 only

B. 1, 3 and 5 only

C. 1, 2 and 4 only

D. 1, 2 and 5 only

AM > GM > HM

$\sqrt[n]{x_1 \times x_2 \times x_3 \cdots x_n}$

Answer: C

Solution:

Statement 1: Correct: For any set of positive numbers: GM≤AM and equality occurs only when all values are identical. Since the multipliers are different, the Geometric Mean must be lower than the Arithmetic Mean.

Statement 2: Correct: GM is calculated as: $GM=(x1x2...xn)^{1/n}$. If any observation = 0, the product becomes 0, therefore: $GM=0^{1/n}=0$. Hence, GM becomes zero.

Statement 3: Incorrect Geometric Mean cannot be computed when any observation is negative. The product may become negative and the nth root of a negative number is not defined in real numbers (for even roots). Therefore GM is not calculated for negative values.

Statement 4: Correct One of the most important financial interpretations: GM=Compounded Average Growth Rate. It represents a single constant growth factor which, when applied repeatedly for the same number of periods, gives the same cumulative growth as the actual sequence of multipliers. This is why GM is used in portfolio returns and CAGR calculations.

Statement 5: Incorrect: Arithmetic averaging of percentage growth rates ignores compounding. Thus: AM(growth rates)≠true compounded growth. Only Geometric Mean correctly captures compounding effects in finance and investment analysis.

3. A financial analyst is comparing the performance of three portfolio managers who generated returns at different annualized rates during varying time segments. Since the returns are expressed as rates, the analyst decides to use an appropriate measure of central tendency.

The analyst also wants to check certain theoretical relationships among Arithmetic Mean (AM), Geometric Mean (GM), and Harmonic Mean (HM) and applies them to evaluate consistency in the data.

$AM \times HM = (GM)^2$

Based on the analyst's understanding, consider the following statements:

1. Harmonic Mean is the reciprocal of the arithmetic mean of reciprocals of the observations and is generally suitable when observations represent ratios such as rates of return or speed.
2. When observations represent ratios of two variables with different measurement units, Harmonic Mean generally provides a more meaningful average than Arithmetic Mean.
3. If the Arithmetic Mean of two numbers is 16 and their Geometric Mean is 8, the Harmonic Mean of the two numbers will be 4.
4. For any positive dataset, the general inequality relationship among the three means is: Arithmetic Mean ≥ Geometric Mean ≥ Harmonic Mean.
5. If the returns of three investments are 10%, 20%, and 40%, the Harmonic Mean of these returns will always be greater than their Arithmetic Mean.

$AM > GM > HM$

$HM = \sum \frac{1}{n}$

$(16)(HM) = (8)^2$

$16 \times HM = 64$

$HM = \frac{64}{16} = 4$

Which of the following is correct?

A. 1, 2, 3 and 4 only
B. 1, 2 and 4 only
C. 2, 3 and 5 only
D. 1, 3, 4 and 5 only

Answer: A

Solution:

Statement 1 – Correct: Harmonic Mean is mathematically defined as: $HM = \frac{n}{\sum \frac{1}{x_i}}$. Thus, it represents the reciprocal of the arithmetic mean of reciprocals of the observations. It is particularly useful for rates, ratios, and speeds

Statement 2 – Correct: Harmonic Mean is most appropriate when:

- Data consists of ratios or rates
- Numerator and denominator have different measurement units

Arithmetic Mean may distort such averages, whereas HM adjusts for the reciprocal relationship.

Statement 3 – Correct Given: Arithmetic Mean (AM) = 16; Geometric Mean (GM) = 8

Relationship among means: $AM \times HM = (GM)^2$

Substituting values: $16 \times HM = 8^2$; $M = 4$

Statement 4 – Correct: For any positive dataset, the inequality relationship is: $AM \geq GM \geq HM$ Equality occurs only when all observations are identical.

Statement 5 – Incorrect: For the values 10, 20, 40: Arithmetic Mean: $AM = (10+20+40)/3 = 23.33$. Harmonic Mean: $HM = 3/\ 0.175 = 17.14$. $HM < AM$

Harmonic Mean cannot exceed Arithmetic Mean for positive numbers.

4. A financial analyst is examining the distribution of daily transaction values (in ₹ thousand) processed by a fintech platform during a particular week. The analyst compiles the following grouped frequency distribution:

Transaction Value (₹ thousand)	Number of Transactions
0–10	6
10–20	14
20–30	24
30–40	x
40–50	16
50–60	10

Preceding CF, frequency, GF: 6, 20, 44, 44 + x, 60 + x, 70 + x

$L + \left[\frac{\frac{N}{2} - CF}{f}\right] (UL - LL)$

$32 = 30 + \left[\frac{35 + x - 44}{x}\right] 10$

$\frac{70 + x}{2} = 35 + x$

$32 = 30 + \left[\frac{-9 + x}{x}\right] 10$

$x = 30$

It is known that:

- The median transaction value is ₹32 thousand.
- The class width is uniform.
- The lower boundary of the median class is considered after converting to continuous intervals.
- Total observations are 70 + x.

Using the concept of median, quartiles, cumulative frequency, and interquartile range, consider the following statements:

1. The median class is the class interval 30–40.
2. The value of the missing frequency x equals 20.
3. If the correct value of x is used, the first quartile (Q_1) will lie in the 20–30 class interval.
4. The interquartile range (IQR) can be computed even if the mean of the distribution is unknown.
5. The median would remain unchanged if extremely large observations were added to the highest class interval.

Which of the following is correct?

A. 1, 3 and 4 only
B. 1, 2, 3 and 4 only
C. 1, 3, 4 and 5 only
D. 2, 3 and 5 only

Answer: C

Solution:

Statement 1: Correct: Median = 32. The class containing 32 is 30–40, therefore this is the median class.

Statement 2: Incorrect: First compute cumulative frequency:

Class	f	cf
0–10	6	6
10–20	14	20
20–30	24	44
30–40	x	44 + x
40–50	16	60 + x
50–60	10	70 + x

Total frequency N = 70 + x. Median formula for grouped data: $Median = l + \left(\frac{N/2 - cf}{f}\right) \times h$. Where l=30, cf=44, f=x, h=10. x=30. Thus x ≠ 20. Statement 2 is incorrect.

Statement 3: Correct: Now total observations: N=70+30=100. First quartile position: Q_1=N/4=25.

Cumulative frequencies:

Class	cf
0–10	6
10–20	20
20–30	44

Since 25 lies between 20 and 44, Q_1 lies in the 20–30 class.

Statement 4: Correct: Interquartile Range: IQR=Q3−Q1. Both Q_1 and Q_3 depend on cumulative frequency positions, not on the mean. Therefore the mean is not required.

Statement 5: Correct: Median is a positional measure and depends only on the middle observation. Adding extreme values at the upper end changes the mean significantly but usually does not affect the median, unless the number of observations shifts the middle position drastically.

Class

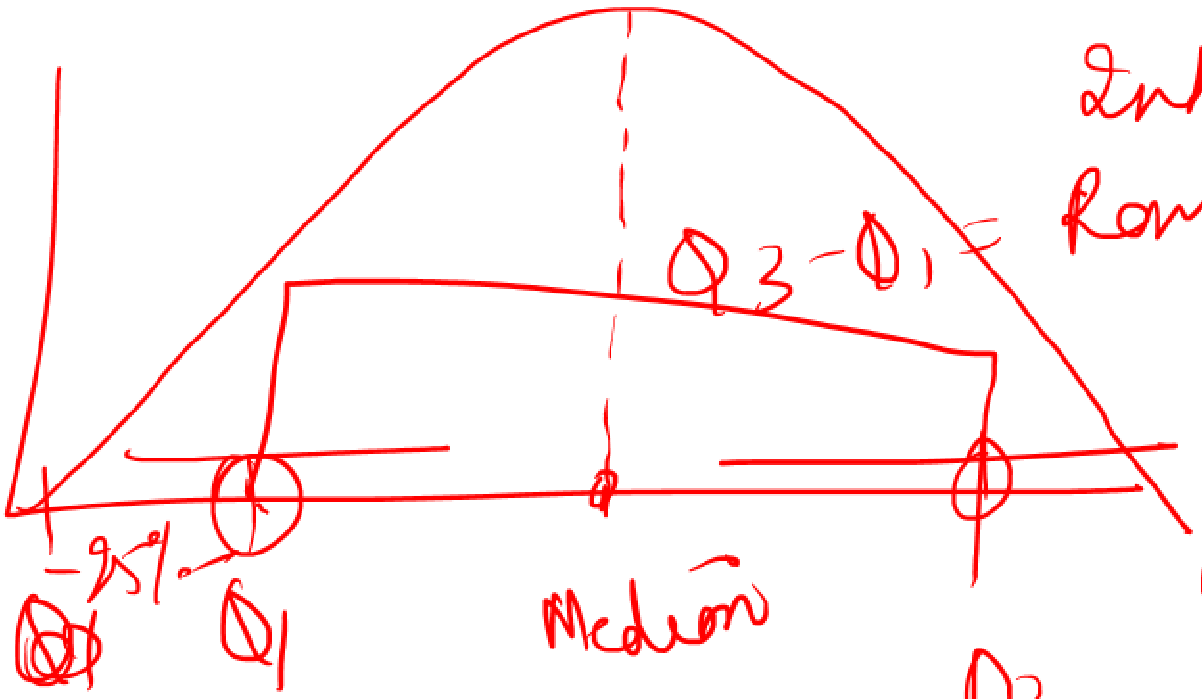
Median $\frac{N}{2}$

$Q_1 = \frac{N}{4}$

$Q_2 = \frac{2N}{4}$ = Median $= \frac{N}{2}$

$Q_3 = \frac{3N}{4}$

$\frac{100}{4} = 25$



Inter Quartile Range

$Q_3 - Q_1 =$ Range

$\frac{N}{4} = Q_1$

$\frac{N}{2} = \frac{2N}{4} = Q_2$

$\frac{3N}{4} = Q_3$

50%

$\frac{N}{2} = Q_2$

75%

$L_1 + \left[\frac{\frac{N}{2} - cf}{f}\right] \times h^{L_2 - L_1} = 32$

5. A data analyst in a financial institution is examining the distribution of transaction processing times (in seconds) recorded over a specific period. The following grouped frequency distribution was obtained:

Processing Time (seconds)	Number of Transactions
10–20	12
20–30	18 f_0
30–40	35 f_1
40–50	30 f_2
50–60	15

$$\text{Mode} = L_1 + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right] \times h \qquad L_2 - L_1$$

$$30 + \left[\frac{35 - 18}{70 - 18 - 30}\right] \times 10$$

During the analysis, the analyst also computed the mean processing time as 36 seconds and estimated the median processing time as 37 seconds.

$$\text{Mean} = 3\,\text{Median} - 2\,\text{Mode}$$

$$2\,\text{Mode} = 3\,\text{Med} - 1\,\text{Mean}$$

Based on the above information, evaluate the following statements:

1. The modal class of the distribution is 30–40 seconds.
2. The value of mode calculated using the grouped data formula will lie between 33 and 36 seconds.
3. If the empirical relation between mean, median and mode is used, the approximate mode will be 39 seconds.
4. If the frequency of the class 40–50 increases sufficiently to exceed the frequency of 30–40, the modal class would shift even if other class frequencies remain unchanged.
5. Mode is always a reliable representative measure of central tendency for grouped distributions because it is based on all observations.

$$2\,\text{Mode} = 3(37) - 36$$

$$\text{Mode} = \frac{3 \times 37 - 36}{2}$$

A. 1, 2 and 4 only
B. 1, 3 and 4 only
C. 2, 3 and 5 only
D. 1, 2, 3 and 4 only

Answer: B

Solution:

Statement 1: Correct: The modal class is the class with the highest frequency. Frequencies: 10–20: 12; 20–30: 18; 30–40: 35 (highest); 40–50: 30; 50–60: 15. Therefore, 30–40 is the modal class.

Statement 2: Incorrect: Mode formula for grouped data: $$Mode = l + \frac{(f_1 - f_0)}{(2f_1 - f_0 - f_2)} \times h$$

Where:

- l=30; f1=35; f0=18; f2=30; h=10. Mode =30+7.73 = 37.73

Statement 3: Correct: Empirical relation: Mode=3(Median)−2(Mean); Mode=3(37)−2(36)Mode = 3(37) - 2(36)Mode=3(37)−2(36) =111−72=39.

Statement 4: Correct: Modal class depends only on maximum frequency. If frequency of 40–50 becomes greater than 35, that class becomes the new modal class, even if other frequencies remain unchanged.

Statement 5: Incorrect: Mode has several limitations:

- It does not use all observations.
- It is not suitable for algebraic treatment.
- It may be ill-defined in some distributions.

Hence it cannot always be considered a reliable representative measure.

6. A risk analytics team in a commercial bank is analysing the variability of daily profit figures (₹ in lakhs) generated by a proprietary trading desk during a particular week. The following observations are recorded: 12, 18, 16, 25, 14, 20, 10, 30

The Chief Risk Officer asks the analyst to evaluate the dispersion of the dataset and comment on the usefulness and limitations of the Range and related measures while preparing a quick risk note for the management.

Based on the above situation, examine the following statements:

High − low = 30 − 10 = 20

1. The Range of the dataset is 20, and the Coefficient of Range equals 0.50.
2. If the extreme observations 10 and 30 are removed from the dataset, the Range decreases but the mean of the remaining observations must necessarily remain unchanged.
3. The Range is sensitive to extreme values because it is determined only by the maximum and minimum observations, ignoring the remaining values in the dataset.
4. Even though Range does not use all observations, it is still frequently used in quality control, share price fluctuations, and weather forecasting because it provides a quick indication of variability.
5. A good measure of dispersion should ideally be rigidly defined, based on all observations, easy to compute, capable of algebraic treatment, and relatively stable against sampling fluctuations, conditions which the Range satisfies completely.

Highly sensitive

$$25 - 12 = 13$$

Which of the following is correct?

A. Only 1, 3, 4 and 5 are correct

B. Only 1, 2, 3 and 4 are correct

C. Only 2, 4 and 5 are correct

D. Only 1, 3 and 4 are correct

Answer: D

Solution:

Statement 1 — Correct: Maximum value = 30; Minimum value = 10 Range (R) = Max−Min=30−10=20. Coefficient of Range Max−Min / Max + Min = 0.5

Statement 2 — Incorrect: Removing 10 and 30 changes the dataset: Remaining values: 12, 18, 16, 25, 14, 20

The mean changes because extreme values contribute to the overall average. Hence the mean cannot remain unchanged.

Statement 3 — Correct Range uses only two observations:

- Maximum
- Minimum

All other observations are ignored, making it highly sensitive to extreme values.

Statement 4 — Correct: Despite limitations, Range is widely used where quick spread estimation is needed, such as:

- Quality control monitoring
- Share price fluctuations
- Weather forecasting

This is because it is very simple and quick to compute.

Statement 5 — Incorrect: Characteristics of a good dispersion measure include:

- Rigid definition; Based on all observations; Easy to calculate; Algebraically treatable; Stable against sampling fluctuations

Range fails several of these conditions:

- Not based on all observations; Highly sensitive to sampling fluctuations; Poor algebraic properties

Hence it does not satisfy all characteristics of a good dispersion measure.

7. A financial analyst is examining the dispersion in income data of employees in a large organization. The following grouped data represents the monthly income (in ₹ thousands) of 100 employees:

Income Class (₹ '000)	Frequency
20–30	10
30–40	18
40–50	32
50–60	25
60–70	15

After performing calculations, the analyst obtained the following results:

Mean income = ₹46 thousand

- $\Sigma f|x - \bar{x}| = 980$
- First quartile (Q_1) = 35
- Third quartile (Q_3) = 57Which of the following statements are correct?

1. The Mean Deviation about Mean for the distribution is 9.8.
2. The Coefficient of Mean Deviation about Mean is approximately 0.213.
3. The Quartile Deviation for the distribution is 11.
4. The Coefficient of Quartile Deviation is approximately 0.239.
5. If a few extremely high incomes are added to the dataset, Quartile Deviation will remain largely unaffected whereas Mean Deviation will necessarily remain unchanged.

A. 1, 2 and 3 only
B. 1, 2, 3 and 4 only
C. 2, 3 and 5 only
D. 1, 2 and 5 only

$QD = \frac{Q_3 - Q_1}{2}$

$x - \bar{x}$: 25−46, 35−46, 45−46, 55−46, 65−46

$f|x - \bar{x}|$

$\frac{\Sigma f|x - \bar{x}|}{\Sigma f}$

$\frac{980}{100}$

$\frac{9.8}{46}$

$\frac{Q_3 - Q_1}{Q_3 + Q_1}$ — Range

Answer: B

Solution:

Statement 1: Correct: Mean Deviation about Mean (grouped data): $MD = \frac{\sum f|x - \bar{x}|}{\sum f}$ = 9.8

Statement 2: Correct: Coefficient of Mean Deviation (about mean): Coefficient of MD = MD/ Mean = 9.8/46 = 0.213

Statement 3: Correct: Quartile Deviation: QD = Q3-Q1/2 = 22/2 = 11

Statement 4: Correct: Coefficient of Quartile Deviation: Q3-Q1/Q3+Q1 = 22/92 =0.239

Statement 5: Incorrect: Quartile deviation depends only on Q_1 and Q_3, which are positional measures and hence not influenced by extreme values.

Mean deviation uses all observations, therefore extreme values will change the mean and deviations, affecting MD.

8. A risk analytics team of a financial institution is evaluating the variability and distributional characteristics of monthly returns generated by two algorithmic trading strategies, Strategy A and Strategy B.

For Strategy A, five monthly returns (%) were recorded as: 2, 3, 7, 8, 10.

For Strategy B, the returns were summarized in grouped form as follows:

Return Interval (%)	Frequency
25–30	30
30–35	23
35–40	20
40–45	14
45–50	10
50–55	3

100

$$\sigma = \sqrt{\frac{\Sigma fx^2}{N} - \left(\frac{\Sigma fx}{N}\right)^2}$$

$\Sigma fx = 3550$

$\Sigma fx^2 = 131225$

- The analytics team also observed the following statistical properties:
- Mean return of Strategy A = 6%.
- Standard deviation of Strategy A ≈ 3.03.
- Total frequency for Strategy B = 100.
- Σfx = 3550 and Σfx² = 131225 for Strategy B.

C.V

$$\frac{\sigma}{\bar{x}} \times 100 = \frac{3.03}{6} \times 100 =$$

Additionally, the team evaluates the shape of a third return distribution (Strategy C) where the third central moment equals zero, while the fourth central moment is smaller relative to the squared second central moment.

Which of the following statements are correct?

$$\sqrt{\frac{131225}{100} - \left(\frac{3550}{100}\right)^2}$$

1. The Coefficient of Variation of Strategy A is approximately 50.5%, indicating a relatively high variability of returns compared to its mean.
2. Using the grouped data formula, the standard deviation of Strategy B is approximately 7.21, implying greater dispersion than Strategy A.
3. If Strategy C has a third central moment equal to zero, the distribution must be perfectly symmetrical around the mean.
4. If the value of β_2 (kurtosis) for Strategy C is less than zero, the distribution would be classified as platykurtic, indicating thinner tails compared to a normal distribution.
5. Standard deviation gives greater weight to extreme observations because deviations from the mean are squared during calculation.

A. 1, 2, 3 and 5 only

B. 1, 2 and 5 only

C. 1, 2, 4 and 5 only

D. 2, 3, 4 and 5 only

Answer: C

Solution:

Statement 1: Correct: Coefficient of Variation: $CV = \frac{\sigma}{\bar{x}} \times 100$

For Strategy A:CV=3.03/6×100=50.5%

This shows high variability relative to the mean return.

Statement 2: Correct

For grouped data: $\sigma = \sqrt{\frac{\Sigma f x^2}{N} - \left(\frac{\Sigma f x}{N}\right)^2}$ ≈7.21

Dispersion in Strategy B returns is larger than Strategy A.

Statement 3: Incorrect

A third central moment equal to zero implies no skewness, but it does not guarantee perfect symmetry in every statistical situation. It indicates absence of skewness measure but symmetry cannot always be strictly concluded in empirical distributions.

Statement 4: Correct

Kurtosis measure: $\beta_2 = \frac{\mu_4}{\mu_2^2}$

Interpretation:

- $\beta_2 = 0 \rightarrow$ Mesokurtic
- $\beta_2 > 0 \rightarrow$ Leptokurtic
- $\beta_2 < 0 \rightarrow$ Platykurtic

A negative β_2 implies a flatter peak and thinner tails compared to the normal distribution.

Statement 5: Correct: Standard deviation squares deviations: $(x - \bar{x})^2$ Therefore extreme values receive disproportionately larger weight, which is one of the known limitations of SD.

9. Which of the following statements are TRUE regarding various random sampling methods?

1. In **simple random sampling**, each possible sample has an equal probability of being selected, and so does each individual item in the population.
2. **Systematic sampling** ensures both equal chances for each item and each possible combination (sample) to be selected.
3. **Stratified sampling** is best used when the population is divided into internally homogenous but mutually heterogeneous groups.
4. In **sampling with replacement**, the same element may appear more than once in the sample.
5. **Cluster sampling** is more suitable when the population groups are internally diverse but structurally similar to each other.
6. Systematic sampling cannot introduce any bias or sampling error, regardless of the data's pattern.
7. In statistical inference, all sampling methods are equivalent and treated as identical to simple random sampling.

A. Statements 2, 3, 4, and 5 are true
B. Statements 1, 2, 3, 4, and 6 are true
C. Statements 1, 3, 4, and 5 are true
D. Statements 1, 4, 5 and 6 are true

Answer: C

Explanation:

True – Simple random sampling gives every individual and every possible sample combination an equal chance of selection. This is its defining feature.

False – In systematic sampling, **each item** may have an equal chance of selection, but **each combination/sample** does **not** have equal probability.

True – Stratified sampling is used when the population has natural subdivisions that are **internally similar** but **different from each other**.

True – Sampling **with replacement** allows the same individual to appear more than once in the final sample.

10. Which of the following statements regarding **sampling distributions and standard error** are correct?

1. The standard error represents the standard deviation of a sample from a population.
2. The mean of a sampling distribution is equal to the population mean if the sampling is unbiased.
3. A lower standard error implies a more accurate estimate of the population parameter.
4. Sampling distributions allow us to understand the variability of a statistic due to chance.
5. Larger samples always reduce standard error to zero, regardless of population variability.

Choose the correct option:

A. Only 2, 3, and 4
B. Only 1, 2, and 5
C. Only 3 and 5
D. Only 1, 2, 3, and 4

Answer: A

Statement 1: **Incorrect** – The **standard error** is **not** the standard deviation of *a sample*; it is the **standard deviation of the sampling distribution** of a sample statistic (e.g., mean or proportion). It captures **how much the sample means (not the data values)** deviate from the population mean.

Statement 2: **Correct** – If sampling is **unbiased**, the mean of the sampling distribution (i.e., the average of all sample means) is equal to the **population mean**. This is a fundamental property of unbiased estimators.

Statement 3: **Correct** – A **smaller standard error** means there is **less variability** among sample statistics. Therefore, the estimate of the population parameter is **more accurate** and **more reliable**.

Statement 4: **Correct** – **Sampling distributions** help us understand the **chance error** or variability that arises just because we're sampling, not surveying the whole population. It's key to inferential statistics.

Statement 5: **Incorrect** – **Larger samples reduce** the **standard error**, but **never to zero** unless the entire population is surveyed. Moreover, **population variability** still influences standard error.

11. Which of the following statements are TRUE with respect to the sampling distribution of the
sample mean when the population is normally distributed?

1. The mean of the sampling distribution is equal to the population mean.
2. The standard deviation of the sampling distribution becomes zero if the sample size equals the population size.
3. The sampling distribution has a smaller spread than the population itself.
4. If population standard deviation is o and sample size is n, the standard deviation of the sample mean is o/Vn.
5. Increasing the sample size increases the standard deviation of the sampling distribution

A. Only 2, 3, and 4
B. Only 1, 2, and 5
C. Only 1, 3 and 4
D. Only 1, 2, 3, and 4

Answer: C

Statement 1: True - The mean of the sampling distribution equals the population mean, denoted as **= ^**.

Statement 2: False - The standard deviation doesn't become zero even if the sample equals population; instead, finite population correction is applied.

Statement 3: True - Sample means exhibit **less variability** than individual observations, hence smaller spread.

Statement 4: True - This is the formula for **standard error of the mean**: **Ox = o / Vn**.

Statement 5: False - Increasing sample size **reduces** the standard deviation (spread) of the sampling distribution.

12. **Which of the following statements related to the Central Limit Theorem (CLT) and its applications is/are correct?**

1. As per the Central Limit Theorem, the sampling distribution of the sample mean approaches a normal distribution only if the population distribution is normal and the sample size exceeds 30.

2. The standard error of the mean decreases as the sample size increases, due to its inverse relationship with the square root of the sample size.

3. Even if the population distribution is highly skewed, the distribution of sample means becomes approximately normal as the sample size grows.

4. The CLT allows us to use sample statistics to make inferences about population parameters, even without knowing the shape of the population distribution.

5. Increasing the sample size always proportionally increases the precision of the estimate, regardless of cost-benefit considerations.

A. Only 2, 3, and 4

B. Only 1, 2, and 5

C. Only 1, 3 and 4

D. Only 1, 2, 3, and 4

Answer: A

Statement 1: Incorrect - The CLT does **not** require the population distribution to be normal. It states that **regardless of the population's distribution**, the **sampling distribution** of the sample mean will **tend toward normality** as the sample size increases (typically **n > 30** is considered sufficient, but even smaller samples can suffice in practice).

Statement 2: Correct - The **standard error of the mean (sx)** is calculated as **o/Vn**. Therefore, **as sample size increases**, the denominator becomes larger, and the **standard error becomes smaller**,
leading to tighter clustering of sample means around the population mean.

Statement 3: Correct - A powerful implication of CLT is that it **works even when the population distribution is skewed or not normal**. As **n increases**, the **sampling distribution** of the mean becomes **approximately normal**.

Statement 4: Correct - The **core value of CLT** is that it allows **statisticians to make inferences** using the **normal distribution**, even if the population distribution is unknown. This is critical in real-world applications where population data is often not available or not normally distributed.

Statement 5: Incorrect - While a **larger sample size increases precision**, the **return on precision diminishes** due to the square root relationship. Additionally, **practical limitations like cost** must be considered; beyond a point, the **benefit may not justify** the additional effort or resources required for sampling.

13. **Which of the following statements is/are TRUE regarding the finite population multiplier and its effect on the standard error of the mean?**

1. When the population size is significantly larger than the sample size (i.e., sampling fraction is small), the finite population multiplier approaches 1 and has minimal impact on standard error.

2. The finite population multiplier must be used in all situations regardless of the sampling fraction, to ensure statistical accuracy.

3. The standard error of the mean decreases as sample size increases, but the law of diminishing returns applies—doubling the sample size does not cut the standard error by half.

4. The finite population multiplier is calculated using the formula V[(N - n)/(N - 1)], and is applicable when sampling is without replacement from a finite population.

5. If the sample includes more than 5% of the population (i.e., sampling fraction > 0.05), the finite population correction should be considered while computing the standard error.

A. Only 2, 3, and 4

B. Only 1, 3, 4 and 5

C. Only 1, 3 and 4

D. Only 1, 2, 3, and 5

Answer: B

Statement 1 - True: When the sample size is small relative to the population (e.g., 20 out of 1,000), the finite population multiplier is close to 1. In such cases, it doesn't significantly affect the standard error, and the infinite population formula is acceptable.

Statement 2 - False: The finite population multiplier **is not** needed in all scenarios. The generally accepted statistical rule is: if the sampling fraction (n/N) is less than 5%, the multiplier can be ignored.

Statement 3 - True: Increasing sample size reduces the standard error due to its position in the denominator, but **not proportionally**. This illustrates the principle of diminishing returns— larger samples provide more precision but with less marginal gain.

Statement 4 - True: The finite population multiplier is V[(N - n)/(N - 1)] and is applied when we **sample without replacement** from a finite population. It adjusts the standard error downward, reflecting reduced variability.

Statement 5 - True: When more than 5% of the population is sampled (i.e., sampling fraction > 0.05), ignoring the finite population multiplier could lead to **overestimation** of the standard error. Thus, it should be applied.

14. Which of the following statements are **correct** regarding Measures of Central Tendency?

1. Measures of central tendency are used to condense large volumes of data into a single representative value to facilitate analysis and comparison.

2. A good measure of central tendency must always be affected by extreme values to reflect real-life fluctuations accurately.

3. One key objective of central tendency is to simplify complex data sets for clearer interpretation and decision-making.

A. Only 1 and 2
B. Only 1 and 3
C. Only 2 and 3
D. All 1, 2, and 3

Answer: B Explanation:

Statement 1: Correct: Central tendency is used to reduce large amounts of data to a single value that represents the whole, allowing comparative studies and easier conclusions.

Statement 2: Incorrect: A good measure of central tendency should not be unduly affected by extreme values. Sensitivity to outliers makes a measure less representative, which is why one of the requisites is resistance to such distortions.

Statement 3: Correct: One of the main purposes of using averages is to make complex and classified/tabulated data easier to understand and interpret

15. Three different departments in a company have the following average monthly salaries and number of employees:

- Department A: 25 employees, average salary ^48,000 $\bar{x}_1$

N_1 N_2

- Department B: 40 employees, average salary ^52,000 $\bar{x}_2$
- Department C: 35 employees, average salary ^47,000 $\bar{x}_3$

N_3

What is the **combined average monthly salary** of all the employees in the company?

$$\frac{25 \times 48000 + 40 \times 52000 + 35 \times 47000}{25 + 40 + 35}$$

A. ^48,933.33
B. ^50,500.00
C. ^49,250.00
D. ^48,800.00

Answer: C

We use the **combined arithmetic mean formula** extended for three groups:

X combined= $n_1X_1+n_2X_2+n_3X_3$ / $m+n_2+n_3$ Given

- $n_1=25$, $X_1=48000$
- $n_2=40$ $X_2=52000$
- $n_3=35$ $X_3=47000$

Total number of employees: 25+40+35=100

Now the combined mean:

X combined= 4925000/100 = 49250

16. **Which of the following statements regarding the Geometric Mean (GM) are correct?**

1. Geometric Mean is preferred when analyzing data that includes compounding effects over time.

2. GM can be used even when the data set includes zero or negative values.

3. GM is the most suitable mean to calculate average speed over different distances.

HM

4. GM is generally less than the Arithmetic Mean for the same dataset.

5. GM is used in calculating Compounded Annual Growth Rate (CAGR).

A. 1, 2, 3, and 5
B. 2, 3, and 4
C. 1, 3, and 4
D. 1, 4 and 5

Answer: D

- **Statement 1 - Correct:** GM accounts for compounding effects and is ideal for time-series data such as investment returns.

- **Statement 2 - Incorrect:** GM cannot be computed when any value is zero or negative.

- **Statement 3 - Incorrect:** GM is not suitable for calculating speed; Harmonic Mean is more appropriate in such cases.

- **Statement 4 - Correct:** For any data set with variation, GM < AM (except when all values are equal).

- **Statement 5 - Correct:** CAGR calculations in finance are based on the geometric mean.

17. **Which of the following statements accurately describe the Harmonic Mean (HM)?**

1. HM is useful when averaging ratios or rates with different units.
2. HM is the square of the product of AM and GM.
3. HM is appropriate for calculating average speed across multiple journeys of different distances.
4. HM is not affected by extreme values or outliers.
5. HM is computed as the reciprocal of the arithmetic mean of the reciprocals of the values.

A. 1, 3, and 5
B. 2, 3, and 4
C. 1, 3, and 4
D. 1, 4 and 5

Answer - A

Statement 1 - Correct: HM is ideal for data like speed (e.g., km/hr), where values are ratios of different units.

Statement 2 - Incorrect: It's $(GM)^2 = AM \times HM$, not

the other way around.

Statement 3 - Correct: HM is the correct mean for computing average speeds over the same distance or rates in general.

Statement 4 - Incorrect: HM is very sensitive to extremely small values and zero (since it involves reciprocals).

Statement 5 - Correct: That's the definition of Harmonic Mean.

18. Which of the following statements related to *Median* and *Quartiles* are correct?

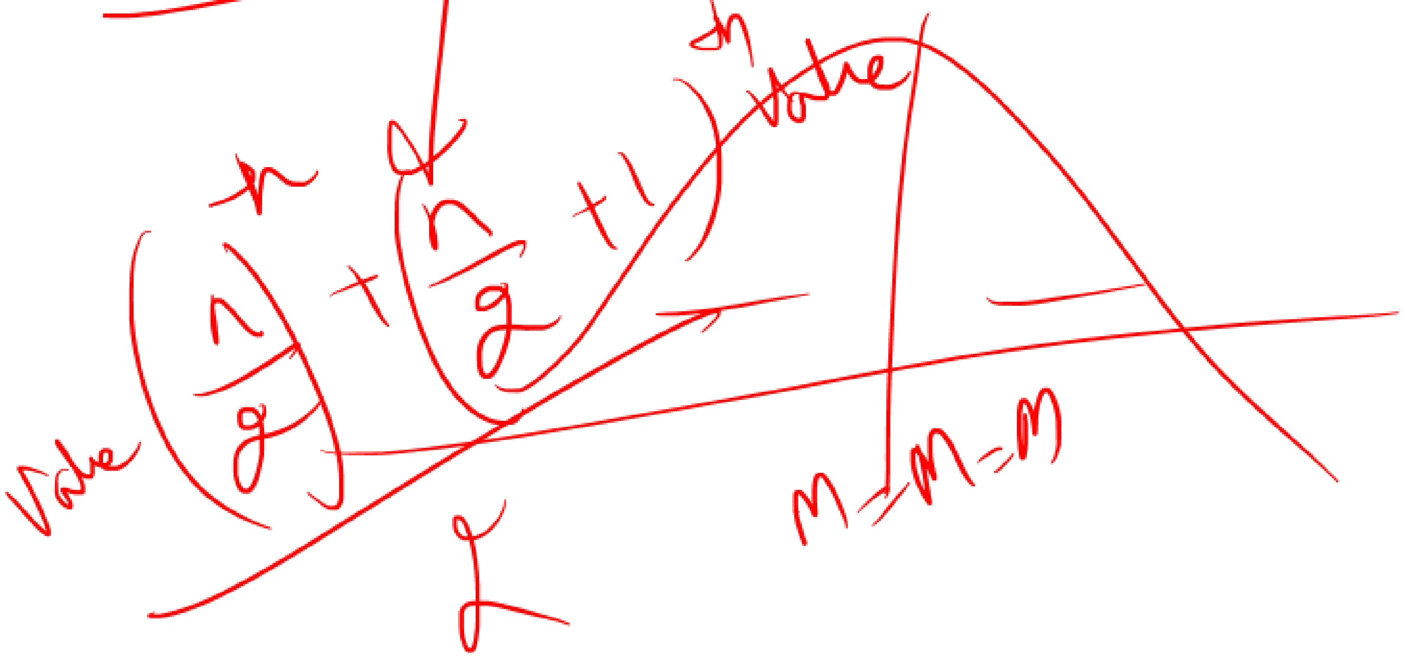
1. The median is always equal to the arithmetic mean in a symmetrical distribution. 2. Quartiles divide the data set into four equal parts using three values: Q1, Q2, and Q3.

3. The interquartile range is calculated as Q3 minus Q1 and represents the spread around the median.

4. For an even number of observations, the median is the average of the two middle values.

5. The median is a positional average and does not depend on the arithmetic calculation of values.

A. 1, 3, and 5
B. 2, 3, and 4
C. 1, 3, and 4
D. 2, 3, 4 and 5



Answer: D

Statement 1: Incorrect - Median equals the arithmetic mean *only* in a symmetric distribution. In skewed distributions, they differ.

Statement 2: Correct - Q1, Q2 (which is the median), and Q3 divide the data into four equal parts.

Statement 3: Correct - The interquartile range (IQR) = Q3 - Q1, and it measures variability around the median.

Statement 4: Correct - For an even number of observations, median is the average of the n/2 and n/2 + 1^{th} term.

Statement 5: Correct - The median is based on the position in an ordered list, not on summing or averaging all values.

19. The following frequency distribution shows the marks obtained by 100 students in a test:

Marks Interval	Frequency	cf
0 – 10	5	5
10 – 20	12	17
20 – 30	18	35
30 – 40	20	55
40 – 50	25	80
50 – 60	10	90
60 – 70	10	100

$\frac{100}{2} = 50$

$30 + \left[\frac{50 - 35}{20}\right][10] =$

What is the median mark of the students?

A. 34.5

B. 35.0

C. 36.25

D. 37.5

Answer: D

Marks Interval	Frequency (f)	Cumulative Frequency (CF)
0 – 10	5	5
10 – 20	12	17
20 – 30	18	35
30 – 40	20	55
40 – 50	25	80
50 – 60	10	90
60 – 70	10	100

Step 2: Find median class

- Total frequency, N=100
- N\2=50
- Median class is the class where CF just exceeds 50 → class **30 – 40**

$$Median = l_1 + \frac{(l_2 - l_1)\left(\frac{N}{2} - cf\right)}{f}$$

l 1 = lower limit of the median class, l 2 = upper limit of the median class f = frequency of the median class, cf = cumulative frequency of the class preceding the median class, N = total frequency.

= 30 + [(40-30)*(50-35)/20] = 37.5

20. Which of the following statements about Mode and its characteristics is/are correct?

1. Mode is the value that appears most frequently in a dataset and can have more than one value in certain distributions.
2. The formula for mode in a grouped frequency distribution considers the frequencies of the modal class, its preceding class, and succeeding class.
3. One merit of mode is that it requires all data points to be known and used in its calculation.
4. A multimodal distribution has more than two modal values that occur with similar frequency.
5. A key limitation of mode is that it is suitable for algebraic manipulations and highly representative of the dataset.

A. Only 1, 2 and 5
B. Only 1, 2 and 4
C. Only 2, 3 and 5
D. Only 1, 3 and 4

Answer: B

Statement 1: Correct. Mode refers to the most frequently occurring value in a dataset. If multiple values share the same highest frequency, the distribution can be bimodal or multimodal. **Statement 2: Correct.** In grouped data, the formula for mode: Mode=$l_1+(f_1-f_0)/(2f_1-f_0-f_2)*h$ uses the frequencies of the modal class (**fi**), the class before it (**f_0**), and the class after it (**f_2**).

Statement 3: Incorrect. One advantage of mode is that **not all data points are needed**—only the class with maximum frequency is necessary.

Statement 4: Correct. A **multimodal** distribution is one where **three or more values** occur with similar high frequencies.
Statement 5: Incorrect. Mode is **not suitable for algebraic treatment**, and it is **not a reliable representative** of the dataset in many cases.

21. Which of the following statements related to measures of dispersion are correct?

1. Measures of dispersion indicate how far individual data points deviate from the central tendency.
2. The Coefficient of Range and Standard Deviation are both relative measures of dispersion.
3. One key characteristic of a good measure of dispersion is that it should be based on all observations.
4. Quartile Deviation and Mean Deviation are considered absolute measures of dispersion.
5. The Coefficient of Variation is an absolute measure used for comparing variability between different datasets.

A. Only 1, 2 and 5

B. Only 1, 2 and 4

C. Only 2, 3 and 5

D. Only 1, 3 and 4

Answer: D

1. **Correct** - Measures of dispersion help assess the degree of spread or variability in data by showing how much individual data points differ from a measure of central tendency (like mean or median).
2. **Incorrect** - The **Standard Deviation** is an **absolute** measure of dispersion, while the **Coefficient of Range** is a **relative** measure. The question mixes two different types.
3. **Correct** - A good measure of dispersion should be **based on all observations** to accurately reflect the overall spread in the dataset.
4. **Correct** - **Quartile Deviation** and **Mean Deviation** are listed among the **four absolute measures of dispersion**, along with Range and Standard Deviation.
5. **Incorrect** - The **Coefficient of Variation** is a **relative** measure of dispersion, not absolute. Itis particularly useful when comparing variability across datasets with different units or means.

22. **Which of the following statements regarding measures of dispersion is/are correct?**

1. Range is not a reliable measure when a dataset includes outliers or extreme values.
2. The coefficient of quartile deviation uses Q1 and Q3 in both numerator and denominator.

$\frac{Q_3-Q_1}{Q_3+Q_1}$

3. Mean deviation is preferred over range in quality control applications due to its algebraic tractability.
4. Quartile deviation is unaffected by extreme values and is useful when data has outliers.
5. The mean deviation is calculated by ignoring algebraic signs of deviations from the mean.

A. Only 1, 2 and 5
B. Only 2, 3 and 4
C. Only 1, 2, 4 and 5
D. All of the above

Answer: C

- **Statement 1 - Correct:** Range only uses maximum and minimum values; a single extreme can distort it.

- **Statement 2 - Correct:** Coefficient of QD = (Q3 - Q1)/(Q3 + Q1), so both quartiles are used in numerator and denominator.

- **Statement 3 - Incorrect:** Mean deviation is not algebraically tractable because it uses absolute values. Hence, it is not preferred in applications requiring algebraic manipulation.

- **Statement 4 - Correct:** QD is not affected by extremes since it uses only middle quartiles (Q1 and Q3).

- **Statement 5 - Correct:** MD uses absolute values (ignores signs), hence the deviations are unsigned.

23. Find the correct pair of range and coefficient of range from the data set below:

Data Set: 24, 31, 18, 29, 20, 22

31
−18

A. Range = 13; Coefficient of Range = 13/49

B. Range = 12; Coefficient of Range = 12/55

C. Range = 13; Coefficient of Range = 13/53

D. Range = 11; Coefficient of Range = 11/51

Answer: A Explanation:

Max value = 31, Min value = 18 Range = 31 - 18 = 13

Coefficient of Range = (Max - Min) / (Max + Min) = (31 - 18) / (31 + 18) = 13 / 49

24. Which of the following statements is/are **correct** with respect to Standard Deviation and Coefficient of Variation?

1. A smaller standard deviation indicates greater consistency and homogeneity in a data set.

2. The coefficient of variation expresses the standard deviation as a percentage of the mean.

3. One of the demerits of standard deviation is that it does not consider extreme values in the data.

A. Only 1 and 2
B. Only 2 and 3
C. Only 1 and 3
D. All 1, 2 and 3

Answer: A

- **Statement 1 - Correct:** A small standard deviation reflects that most of the values are close to the mean, indicating **high consistency and homogeneity**, as clearly highlighted in the text.

- **Statement 2 - Correct:** The **coefficient of variation (CV)** is calculated using the formula:
  CV=o/x"xi00%
  This expresses the standard deviation as a **percentage of the mean**, allowing comparison of variability across data sets with different units or scales.

- **Statement 3 - Incorrect:** This is **false**. In fact, a **demerit** of standard deviation is that it **gives more weight to extreme items**, making it sensitive to outliers—not that it ignores them.

25 .Which of the following statements regarding skewness and kurtosis is/are correct?

1. In a positively skewed distribution, the mean is greater than the median, which in turn is greater than the mode.

2. A distribution with 01 = 0 and 02 = 0 is both symmetrical and mesokurtic.

3. Leptokurtic distributions have fatter tails and a sharper peak than the normal distribution, indicating the presence of more extreme values.

4. A distribution with a negative skew has a longer tail on the left side and the order of central tendency becomes Mode > Median > Mean.

5. The measure of kurtosis 02 is defined using the second and fourth central moments, and 02 < 0 indicates a platykurtic distribution.

A. Only 1, 2 and 5

B. Only 1, 2, 3 and 4

C. Only 1, 2, 4 and 5

D. All of the above

Answer: D **Explanation:**

1. **True** - In **positive skewness**, the **longer tail is on the right**, and the **order becomes Mean > Median > Mode**. This is due to extreme high values pulling the mean to the right.

2. **True** - When **pi = 0**, the distribution is **symmetrical**. When **P2 = 0**, the distribution is **mesokurtic** (same peaked Ness as a normal distribution). If both conditions hold, the distribution is **normal** in shape.

3. **True** - **Leptokurtic distributions** are **peaked sharply** with **fatter tails**, which means there are **more outliers** than in a normal distribution.

4. **True** - In **negative skewness**, the **longer tail is on the left**, and the **order is Mode > Median > Mean**, since low extreme values pull the mean leftward.

5. **True** - **Kurtosis (P2)** is calculated using the **fourth central moment (p4)** and the **square of the second central moment ($p2^2$)**. If **P2 < 0**, the distribution is **platykurtic**, i.e., flatter than normal, with fewer outliers.

JAIIB CAIIB
PREP

UNIT 4

Correlation and Regression

Relation ship

%?

STRUCTURE

Meaning of Correlation

- Correlation measures:
- Degree of relationship between two variables
- Direction of movement between variables
- Example in banking:
- Income & savings
- Interest rate & loan demand
- Inflation & purchasing power
- If one variable changes and another also changes, they are correlated.

Types of Correlation

- **(A) Positive Correlation**
- Both variables move in same direction.
- Example:
- Income ↑ → Savings ↑
- Sales ↑ → Profit ↑
- Coefficient value:
- Between 0 and +1
- Perfect positive correlation:
- $r = +1$

(B) Negative Correlation

- Variables move in opposite direction.
- Example:
- Price ↑ → Demand ↓
- Interest rate ↑ → Borrowing ↓
- Coefficient value:
- Between 0 and −1
- Perfect negative correlation:
- $r = -1$

(C) Zero Correlation

- No relationship between variables.
- Formula:
- $r = 0$

Degree of Correlation

Correlation Coefficient (r)	Interpretation
±1	Perfect correlation
±0.75 to ±1	High correlation
±0.25 to ±0.75	Moderate correlation
0 to ±0.25	Low correlation
0	No correlation

Scatter Diagram Method

- Graphical method of studying correlation.
- Points moving upward → Positive correlation
- Points moving downward → Negative correlation
- Random scatter → No correlation
- Very important conceptual topic.

Karl Pearson's Coefficient of Correlation

- Most important formula for numericals.
- Measures:
- Strength
- Direction of linear relationship
- Formula:
- $r = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sqrt{\sum(x-\bar{x})^2 \sum(y-\bar{y})^2}}$
- Where:
- x, y= variables
- $\bar{x}, \bar{y}$ =means

Properties of Correlation Coefficient

- Important exam points:
- Value lies between −1 and +1
- Unit free measure
- Indicates direction & degree
- Measures only linear relationship

Spearman’s Rank Correlation

- Used when:
- Data is qualitative
- Ranks are given
- Example:
- Ranking employees
- Customer satisfaction ranking
- Formula:
- $r_s = 1 - \frac{6 \sum d^2}{n(n^2-1)}$
- Where:
- d= difference between ranks
- n= number of observations

Meaning of Regression

- Regression studies:
- Cause-and-effect relationship
- Predictive relationship between variables
- Used for forecasting.
- Banking applications:
- Loan default prediction
- Deposit forecasting
- Sales forecasting
- Risk estimation

Difference Between Correlation & Regression

Correlation	Regression
Measures relationship	Predicts values
No dependent variable	Has dependent variable
Symmetrical	Asymmetrical
Degree measurement	Functional relationship

Regression Equations

- Two regression lines:
- **(A) Regression of Y on X**
- Predicts Y from X.
- Formula:
- $Y - \bar{Y} = b_{yx}(X - \bar{X})$

Regression of X on Y

- Predicts X from Y.
- Formula:
- $X - \bar{X} = b_{xy}(Y - \bar{Y})$

11. Regression Coefficients

Important formulas:

Regression coefficient of Y on X

$$b_{yx} = r\frac{\sigma_y}{\sigma_x}$$

Regression coefficient of X on Y

$$b_{xy} = r\frac{\sigma_x}{\sigma_y}$$

12. Relationship Between Correlation & Regression

Very important CAIIB formula:

$$r = \sqrt{b_{xy} \times b_{yx}}$$

Important points:

- Both regression coefficients have same sign
- If one coefficient >1, other must be <1

13. Coefficient of Determination

Denoted by:

$$r^2$$

Measures:

- Proportion of variation explained

Example:

- $r^2 = 0.81$
- Means 81% variation explained.

14. Probable Error (PE)

Used to test reliability of correlation coefficient.

Formula:

$$PE = 0.6745 \times \frac{1-r^2}{\sqrt{n}}$$

Rules:

- If $r < PE$ → correlation not significant
- If $r > 6PE$ → correlation significant

Be Back at 8:15 pm

Tea Break

Important Shortcut Relationships

Positive Skew Type Relation

$$Mean > Median > Mode$$

Correlation & Regression Relation

$$r = \sqrt{b_{xy}b_{yx}}$$

1. A research analyst is examining whether years of education can be used to predict annual income for a group of individuals. He collects data for 50 individuals and constructs a scatter diagram. On analysis, he observes the following:

- In Dataset A, points are closely clustered around an upward sloping straight line.
- In Dataset B, points are widely scattered but still show an upward trend.
- In Dataset C, points initially decrease and then increase forming a curve.
- In Dataset D, points are randomly scattered with no visible pattern.
- The analyst attempts to draw a line of best fit to estimate income from education.

+1

$y = a + bx$

Based on the above observations, which of the following statements are correct?

1. Dataset A indicates a strong positive linear relationship and will yield more reliable predictions compared to Dataset B.
2. Dataset B, despite showing a positive trend, will have a lower correlation coefficient than Dataset A due to higher dispersion of points.
3. Dataset C represents a curvilinear relationship; hence, applying simple linear regression would not provide an accurate predictive model.
4. Dataset D implies zero correlation, and therefore regression analysis cannot be meaningfully applied for prediction.
5. If the correlation coefficient (r) for Dataset A is close to +1, it implies perfect prediction with no estimation error in regression.

(A) Only 1, 2, 3 and 4 are correct
(B) Only 1, 2 and 3 are correct
(C) Only 1, 2, 3 and 5 are correct
(D) Only 2, 3, 4 and 5 are correct

Answer: A

Solution:

Statement 1 — Correct: When points lie close to a straight upward sloping line, it indicates a strong positive linear relationship. Such data produces better predictive accuracy because deviations from the regression line are minimal.

Statement 2 — Correct: Dataset B has more scatter, meaning greater variability around the line. This reduces the strength of correlation (r), even though the direction remains positive.

Statement 3 — Correct: Dataset C shows a curvilinear (non-linear) relationship. Linear regression assumes linearity, so applying it here would lead to biased or poor predictions.

Statement 4 — Correct: Dataset D shows no discernible pattern, implying no correlation ($r \approx 0$). In such a case, regression analysis has no predictive power.

Statement 5 — Incorrect
Even if $r = +1$, it indicates perfect linear correlation, but:

- In practical datasets, perfect correlation is extremely rare.
- More importantly, perfect correlation does not automatically imply zero prediction error unless all points lie exactly on the regression line, which is a theoretical situation.

Thus, the statement overgeneralizes and is conceptually flawed.

2. A risk analytics team in a bank is evaluating the relationship between daily changes in interest rates (X) and bond portfolio returns (Y) over a period of 6 days. Due to system constraints, only the following aggregated data is available:

Sum of X = 30
Sum of Y = 42
Sum of X^2 = 190
Sum of Y^2 = 322
Sum of XY = 248 — 260
Number of observations (N) = 6

$$r = \frac{N\sum XY - (\sum X)(\sum Y)}{\sqrt{\left(N\sum X^2 - (\sum X)^2\right)\left(N\sum Y^2 - (\sum Y)^2\right)}}$$

$$\frac{6 \times 248 - [30 \times 42]}{\sqrt{\left(6 \times 190 - (30)^2\right)\left(6 \times 322 - (42)^2\right)}}$$

$$\boxed{r = 1.135}$$

The analyst computes the correlation coefficient using the computational formula.

1. The covariance between X and Y is positive.
2. The correlation coefficient (r) is greater than 0.8.
3. If all values of X are increased by a constant, the correlation coefficient will change.
4. The standard deviation of X is less than the standard deviation of Y.
5. If Sum of XY were 260 instead of 248, the correlation coefficient would increase.

Which of the statements given above are correct?

A. 1, 3, 4 and 5 only
B. 1, 2, 4 and 5 only
C. 2, 3, 4 and 5 only
D. 1, 2 and 5 only

Answer: D

Solution:

$$r = \frac{N\sum XY - (\sum X)(\sum Y)}{\sqrt{[N\sum X^2 - (\sum X)^2][N\sum Y^2 - (\sum Y)^2]}}$$

Numerator = 6×248 − (30×42) = 1488 − 1260 = 228 → positive → covariance is positive → Statement 1 is correct.

Denominator:

For X: 6×190 − 30^2 = 1140 − 900 = 240

For Y: 6×322 − 42^2 = 1932 − 1764 = 168

r = 228 / √(240×168) ≈ 228 / 200.8 ≈ 1.135

Thus r > 0.8 → Statement 2 is correct (though value >1 indicates inconsistency in data).

Adding a constant to X does not change correlation → Statement 3 is incorrect.

Standard deviations:

σX = √(240/6) = √40 ≈ 6.32

σY = √(168/6) = √28 ≈ 5.29

Thus $\sigma X > \sigma Y$ → Statement 4 is incorrect.

If Sum of XY increases, numerator increases → r increases → Statement 5 is correct.

3. A risk analytics team in a bank is studying the relationship between credit score (X) and probability of default (Y) for a portfolio of borrowers. A scatter plot suggests a linear relationship. The team applies the least squares regression technique and estimates the regression line:

$Y = a + bX$

They observe that the covariance between X and Y is negative, and the regression line passes close to most data points with roughly equal dispersion above and below the line.

Based on this context, evaluate the following statements:

1. A negative covariance between X and Y implies that the estimated slope bbb of the regression line must also be negative.
2. The intercept aaa represents the expected probability of default when the credit score is zero, even if such a value is outside the observed data range.
3. The least squares method ensures that the sum of absolute deviations between actual and predicted Y values is minimized.
4. If all observed points lie exactly on the regression line, the correlation coefficient between X and Y would be either +1 or −1.
5. Even if the regression line fits well, it is not necessary that the residuals (errors) sum to zero unless explicitly constrained.

Which of the statements given above are not correct?

A. 1, 2 and 4 only
B. 1, 2, 3 and 4 only
C. 3, and 5 only
D. 1, 3, and 5 only

Answer: A

Solution:

Statement 1: Correct: Slope b = Cov(X,Y) / Var(X). Since variance is always positive, the sign of b depends on covariance. Negative covariance implies negative slope.

Statement 2: Correct: Intercept a represents the value of Y when X = 0. This holds even if X = 0 is outside the observed data range.

Statement 3: Incorrect: Least squares minimizes the sum of squared deviations, not absolute deviations.

Statement 4: Correct: If all points lie exactly on the regression line, there is a perfect linear relationship. Hence, correlation coefficient is either +1 or -1.

Statement 5: Incorrect: In least squares regression, the sum of residuals is always equal to zero when an intercept is included

4. A financial analyst at an investment firm is studying the relationship between advertising expenditure (X, in Rs. lakhs) and sales revenue (Y, in Rs. lakhs) for a portfolio company. Based on past data of n = 12 observations, the analyst derives the regression equation:

$$\boxed{\hat{Y} = 15 + 2.5X}$$

95% $\boxed{se = \pm 2se}$

Further, the following summary statistics are computed:

Sum of Y = 420
Sum of Y^2 = 18,200
Sum of XY = 9,600

The analyst calculates the Standard Error of Estimate (Se) to assess prediction accuracy and uses it to construct confidence intervals for future predictions.

Based on the above information, which of the following statements are correct?

1. The Standard Error of Estimate (Se) measures the unexplained variation in Y after fitting the regression line.
2. A low value of Se implies that the correlation coefficient between X and Y must be exactly +1.
3. For a predicted sales value of Rs. 65 lakhs when X = 20, the 95% confidence interval for actual sales will be (65 – 2Se, 65 + 2Se).
4. If all observed values of Y lie exactly on the regression line, then Se will be zero and the confidence interval reduces to a single point.
5. Even if Se remains constant, an increase in the number of observations (n) will always reduce the width of the confidence interval.

A. 1, 3 and 4 only
B. 1, 2, 3 and 5 only
C. 2, 4 and 5 only
D. 1, 3, 4 and 5 only

Answer: A

Solution:

Statement 1: Correct: Standard Error of Estimate measures the dispersion of actual values around the regression line. It represents unexplained variation (residual error).

Statement 2: Incorrect: A low Se indicates a good fit but does not imply perfect correlation. Only when Se = 0, correlation becomes perfectly +1 or -1.

Statement 3: Correct: For 95% confidence level, the interval is given by: Predicted value ± 2Se

Hence, (65 – 2Se, 65 + 2Se).

Statement 4: Correct: If all observations lie exactly on the regression line, residuals are zero, hence Se = 0 and the interval collapses to a single value.

Statement 5: Incorrect: In this framework, confidence interval depends on Se. If Se remains unchanged, increasing n does not directly reduce the interval width.

5. A research analyst at a policy think tank examines multiple datasets to study relationships between economic and social indicators. He arrives at the following observations:

- Dataset A shows that all data points lie perfectly on a circular pattern between variables X and Y.
- Dataset B includes data from multiple regions with widely different income levels, education standards, and consumption patterns. The calculated correlation coefficient between variables P and Q is +0.82.
- Dataset C shows a correlation coefficient of +0.95 between variables M (per capita milk consumption) and N (number of mobile phone users) over a 25-year period.
- Dataset D shows a correlation coefficient of −0.88 between variables U and V, both of which are measured consistently across a uniform sample group.
- Dataset E shows a correlation coefficient of +0.60 between variables S and T, where the scatter plot indicates a clear upward sloping straight-line pattern.

Based on the above, which of the following statements are correct?

1. In Dataset A, despite a perfect relationship between X and Y, the correlation coefficient may be zero due to non-linearity.
2. The high correlation observed in Dataset B may be misleading due to lack of homogeneity, leading to spurious correlation.
3. The strong positive correlation in Dataset C necessarily implies that M is a significant cause of changes in N.
4. The correlation coefficient in Dataset D reliably indicates a strong inverse linear relationship, assuming no issues with data consistency.
5. Dataset E is the most appropriate case where correlation analysis can be validly interpreted for linear relationship assessment.

A. Only 1, 2, and 5 are correct
B. Only 1, 2, 4, and 5 are correct
C. Only 2, 3, and 4 are correct
D. Only 1, 3, 4, and 5 are correct

Answer: B

Solution:

Statement 1 – Correct: Correlation coefficient measures only linear relationships. Even if variables have a perfect curvilinear relationship (like a circle), the value of correlation can be zero, which is a major limitation.

Statement 2 – Correct: Dataset B suffers from heterogeneity (different regions with varying characteristics). This can artificially inflate correlation, resulting in spurious correlation, even when no real relationship exists.

Statement 3 – Incorrect: A high correlation (+0.95) does not imply causation. The relationship between milk consumption and mobile usage may be coincidental or driven by a third factor (e.g., economic growth). Correlation only shows association, not cause-effect.

Statement 4 – Correct: A correlation of −0.88 indicates a strong inverse linear relationship, and since the dataset is homogeneous and consistent, the interpretation is statistically reliable (within the limits of correlation analysis).

Statement 5 – Correct: Dataset E shows a clear linear upward trend, making correlation analysis appropriate and meaningful here. This is an ideal case for applying correlation.

6. **Which of the following statements regarding correlation and its computation are correct?**

1. The correlation coefficient (r) is a measure of the curvilinear relationship between two variables.
2. A correlation coefficient of -1 implies a perfect linear relationship with a negative slope.
3. If r = 0, it means that no linear relationship exists between the two variables, although other types of relationships may still be present.
4. The correlation coefficient is affected by the units in which the variables are measured.
5. The value of r lies between -1 and 1, indicating the strength and direction of the linear relationship.

A) 1, 2, 3
B) 1, 3, 4, 5
C) 2, 3, 4
D) 2, 3, 5

Answer: D

Statement 1 - Incorrect: The correlation coefficient measures the **linear** relationship between two variables, **not** curvilinear. A curvilinear relationship may still exist even if r = 0, but r will not capture that.

Statement 2 - Correct: A value of r = -1 indicates a **perfect inverse (negative linear) relationship**. This means all data points lie exactly on a straight line sloping downward.

Statement 3 - Correct: r = 0 means **no linear correlation** exists, but it does **not rule out** a **nonlinear (curvilinear)** relationship.

Statement 4 - Incorrect: The correlation coefficient is **unit-free**; it is based on standardized values and is therefore **not affected by the units** of the original variables.

Statement 5 - Correct: The correlation coefficient **always lies between -1 and 1**. Values close to +1 or -1 indicate **strong** positive or negative linear relationships, respectively, while values near 0 indicate a **weak** linear relationship.

7. Which of the following statements are **true** in the context of regression and correlation analysis?

1. The slope 'b' in the regression equation y=a+bxy = a + bxy=a+bx represents the direction and steepness of the line and is calculated using the ratio of covariance of X and Y to the variance of X.

2. A high correlation coefficient always implies a causal relationship between two variables.

3. A perfectly curved relationship, like points forming a circle, will have a correlation coefficient of zero.

A) 1, and 3
B) 1, and 2
C) 2, and 3
D) 1, 2, and 3

Answer: A

1. **True** - The slope 'b' in the regression equation y=a+bx indicates the direction and rate of change of Y with respect to X. It is calculated using the **covariance of X and Y divided by the variance of X** ($b=cov(X,Y)/s^2X$). This slope tells us whether the line rises or falls and how steep it is.

2. False - A **high correlation** (say, close to 1) does **not imply causation**. The content provides the example of rice consumption and road accidents, which showed high correlation but no logical cause-effect relationship. This is a classic example of **spurious correlation**.

3. **True** - The **correlation coefficient only captures linear relationships**. Even if a relationship is perfectly curvilinear (like all points on a circle), the correlation coefficient will be **zero** because it cannot capture the nonlinear pattern.

Trend
Seasonal
cyclical
unexplained
Irregular

UNIT 5

Time Series

Q1
Q2
Q3
Q4

m1 m2 m3 m12

y1 y2 y3 y3

STRUCTURE

Keywords

TIME SERIES

- **Meaning**
- Data arranged according to time.
- Examples:
- Monthly deposits
- Yearly profits
- GDP growth

Components of Time Series

- Remember:
- TSCI

Letter	Meaning
T	Trend
S	Seasonal
C	Cyclical
I	Irregular

Long period of time

a+b+c

No pattern

Easy Understanding

Trend

Long-term movement.

Seasonal Variation

Occurs regularly.

Example:
Festival shopping increase.

Cyclical Variation

Business cycle changes.

Irregular Variation

Unexpected events.

Example:
COVID, floods.

Trend Measurement Methods

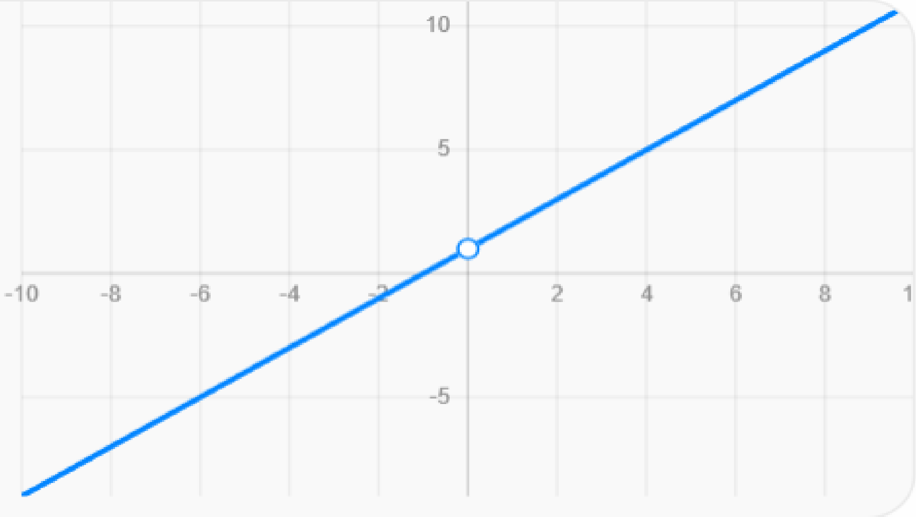
1. Graphic
2. Semi-average
3. Moving average
4. Least square (most important)

Trend equation:

$$Y = a + bX$$

a 1.0

b 1.0



1. A policy analyst in NITI Aayog is studying the time series data of a rapidly growing metropolitan region over a period of 15 years. The dataset includes annual industrial output, quarterly pollution levels, monthly tourist inflows, and sudden shocks due to natural calamities. The analyst observes the following:

- Industrial output shows a long-term upward movement, though with uneven increases.
- Pollution levels exhibit recurring peaks during specific quarters each year.
- Economic activity occasionally rises above and falls below the long-term path, without any fixed periodicity.
- A sudden flood in one year caused a drastic drop in agricultural supply and a spike in prices, which was not anticipated by any prior data trend.
- Despite identifying individual components, isolating them precisely remains challenging due to overlapping effects.

Based on the above, evaluate the following statements:

1. The long-term upward movement in industrial output represents a secular trend, even if the rate of increase is non-uniform.
2. The recurring quarterly peaks in pollution levels can be attributed to cyclical fluctuations since they repeat periodically within a year.
3. The irregular rise and fall of economic activity around the trend line, without fixed periodicity, is characteristic of cyclical variation.
4. The sudden flood-induced disruption in agricultural output and price spike represents irregular variation, which cannot be forecast using time series methods.
5. The presence of multiple overlapping components implies that decomposition of time series into distinct variations is always precise and unambiguous.

A. Only 2, 3, 4 and 5 are correct

B. Only 1, 2, 3 and 4 are correct

C. Only 1, 3, 4 and 5 are correct

D. Only 1, 3 and 4 are correct

Within a yr

Season

Pattern

Fixed

Cycles

Time

over

Not fixed

period of time

Answer: D

Solution:

Statement 1 — Correct: A secular trend captures the long-term direction of data. Even when increases are uneven, the general upward movement over years qualifies as a trend.

Statement 2 — Incorrect: This is a subtle trap. Repetition within a year (quarterly pattern) indicates seasonal variation, not cyclical fluctuation. Cyclical movements occur over multiple years and lack fixed periodicity.

Statement 3 — Correct: Fluctuations around the trend line without a fixed pattern or duration represent cyclical variation, commonly linked to business cycles.

Statement 4 — Correct: Unexpected shocks like floods fall under irregular variation. These are random, non-recurring, and not predictable through historical time series patterns.

Statement 5 — Incorrect: In real-world data, components often overlap, making decomposition complex and approximate—not perfectly precise. This is a key limitation of time series analysis.

2. A leading e-commerce company Amazon is analysing its annual order volumes for the years 2018–2025 to understand long-term growth patterns. Since the number of observations is even, the company applies coding of time and estimates both linear and parabolic trend models. The following results are obtained after coding (mean year = 2021.5):

- $\sum xy = 1680, \sum x^2 = 224, \sum y = 1440, n = 8$
- For parabolic trend: $\sum x^4 = 560, \sum x^2 y = 3920$

$\frac{\sum y}{n} = \frac{1440}{8}$

The analyst wants to forecast order volume for the year 2026 and makes the following statements:

1. The coded value of time for the year 2026 will be x= 9
2. The slope of the linear trend line is b=7.5
3. The intercept a for the linear trend is equal to average of y i.e., a=180
4. The forecast for 2026 using linear trend will be $\hat{y}$ =247.5
5. In parabolic trend estimation, all three coefficients a,b,c must be obtained simultaneously from three equations, and b cannot be estimated independently.
6. If the actual trend exhibits acceleration, the linear forecast for 2026 is likely to underestimate the true value.
7. Coding of time is not required in parabolic trend analysis since higher powers like x^2 and x^4 are already included.

A. 1, 2, 3, 4 and 6 only

B. 1, 2, 4, 5 and 7 only

C. 2, 3, 4, 6 and 7 only

D. 1, 3, 5, 6 and 7 only

$y = 180 + bx$

$= 180 + 7.5(9) = 247.5$

even number coding

$x = 2(T - \text{mean})$

$= 2(2026 - 2021.5)$

$= 9$

Answer: A

Solution:

Statement 1 – Correct: Even-number coding: $x=2(T-\text{mean})=2(2026-2021.5)=9$

Statement 2 – Correct $b=\sum xy/\sum x^2=1680/224=7.5$

Statement 3 – Correct $a=1440/8=180$

Statement 4 – Correct: $\hat{y}=180+(7.5\times 9)=247.5$

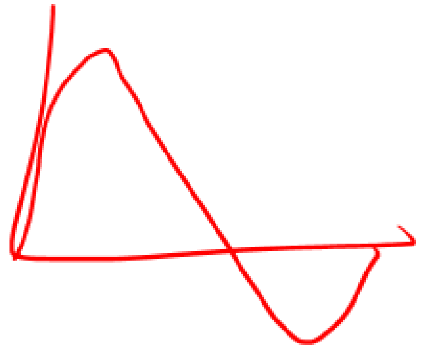
Statement 5 – Incorrect: b is still computed independently using $\sum xy/\sum x^2$; only aaa and ccc require simultaneous equations.

Statement 6 – Correct: Linear trend assumes constant growth: underestimates when growth accelerates.

Statement 7 – Incorrect: Coding is essential to simplify higher power calculations in parabolic models.

3. An economist is analyzing agricultural output data for a country over 10 years. After estimating the secular trend using a least squares method, the economist computes the following:

- Ratio to trend ($Y/\hat{Y} \times 100$) shows fluctuating values between 96% and 105%
- Relative cyclical residuals [$(Y - \hat{Y})/\hat{Y} \times 100$] show alternating positive and negative deviations
- The economist assumes that seasonal variation is negligible
- It is also observed that deviations persist for multiple consecutive years above or below the trend



Based on this, the economist concludes that cyclical patterns dominate the residual variation.

Consider the following statements:

1. If the ratio to trend remains consistently above 100% for multiple years, it necessarily indicates a permanent upward shift in the secular trend rather than cyclical variation.
2. Relative cyclical residual and percentage of trend convey identical information but differ only in scale and interpretation format.
3. The presence of multi-year oscillations above and below the trend line supports the existence of cyclical variation rather than seasonal variation.
4. The residual method assumes that after removing trend, the remaining variation can be fully attributed to cyclical factors if irregular variations are negligible.
5. The residual method used here can be extended reliably for forecasting future cyclical turning points if sufficient historical data is available.

A.Only 2, 3 and 5 are correct
B. Only 1, 3 and 4 are correct
C. Only 2, 3 and 4 are correct
D. Only 2, 3, 4 and 5 are correct

Answer: C

Solution:

Statement 1: Incorrect: Persistent values above 100% indicate that actual values exceed trend, but this does not imply a *permanent shift* in the secular trend. It may still be part of a cyclical upswing. Secular trend requires re-estimation, not inference from residuals alone.

Statement 2: Correct

- Ratio to trend = $(Y/\hat{Y}) \times 100$
- Relative residual = $[(Y - \hat{Y})/\hat{Y}] \times 100$

Both are mathematically related: Relative residual = (Ratio to trend − 100) Thus, they convey the same cyclical information, just expressed differently.

Statement 3: Correct: Cyclical variation spans more than one year, unlike seasonal variation which completes within a year. Multi-year oscillations clearly indicate cyclical behavior.

Statement 4: Correct: Residual method works under the assumption: After removing trend (and seasonality if applicable), remaining variation = cyclical + irregular

If irregular variation is assumed negligible, residual ≈ cyclical variation.

Statement 5: Incorrect
Residual method is backward-looking. It helps in analyzing past cycles but cannot reliably predict future cyclical turning points due to:

- Uncertainty of external shocks
- Non-repetitive nature of cycles
- Absence of deterministic pattern

4. A mid-sized NBFC in Mumbai is analysing its quarterly loan disbursement data for the period 2021–2025 to identify seasonal patterns for credit planning in 2026 using the Ratio to Moving Average Method.

After detailed computation, the following results are obtained:

- Modified seasonal indices (before adjustment): Q1 = 92.80, Q2 = 104.50, Q3 = 118.20, Q4 = 90.70

After applying the adjustment factor, the final seasonal indices are: Q1 = 91.4, Q2 = 103.0, Q3 = 116.5, Q4 = 89.1

The credit team estimates that the deseasonalised trend value for Q4 of 2026 is ₹5,000 crore, while the actual loan disbursement in Q3 of 2024 was ₹5,825 crore.

1. The adjustment factor applied ensures that the aggregate of all quarterly seasonal indices equals 400, thereby preserving proportional seasonal weights
2. The seasonal index of Q4 indicates that loan disbursement in that quarter is, on average, 10.9% lower than the deseasonalised trend.
3. The deseasonalised loan disbursement for Q3 of 2024 is approximately ₹4,500 crore.
4. The forecasted actual loan disbursement for Q4 of 2026, after incorporating seasonal variation, will be approximately ₹4,455 crore.
5. Removing seasonal variation from loan disbursement data isolates the pure trend component, eliminating both cyclical credit cycles and irregular shocks.

A. Only 2, 3 and 4 are correct
B. Only 1, 2 and 4 are correct
C. Only 1, 2 and 3 are correct
D. Only 3, 4 and 5 are correct

$5000 \times \frac{89.1}{100}$ $\frac{5825}{1.165}$

$100 - 89.1 = 10.9$

Cyc + Irreg

Answer: B

Solution:

Statement 1: Correct: For quarterly data, total seasonal indices must equal 400. Adjustment factor ensures normalization without disturbing relative seasonality.

Statement 2: Correct: Q4 index = 89.1. Demand is 10.9% below trend (100 − 89.1)

Statement 3: Incorrect: Deseasonalised value: $5825/1.165 \approx 5000$

Statement 4: Correct: Forecasted disbursement: $5000 \times 89.1/100 = 5000 \times 0.891 = 4455$

Statement 5: Incorrect: Deseasonalisation removes only seasonal variation.
Remaining components = Trend + Cyclical (credit cycles) + Irregular shocks

5. A large financial institution analyzes its quarterly loan disbursement data using the multiplicative time series model (T × S × C × I). The objective is to improve forecasting accuracy for credit planning. The bank first computes seasonal indices and deseasonalises the data. Thereafter, it derives the trend equation:

$Y=18+0.16x$

Using the deseasonalised values and trend estimates, the bank calculates the "percentage of trend" for each quarter. However, significant deviations are observed in certain quarters, even after removing trend and seasonal components. The risk management team suspects the influence of unpredictable macroeconomic shocks such as sudden regulatory changes, geopolitical tensions, and abnormal credit demand surges.

To improve forecasting, the analytics team proposes the use of advanced time series models such as AR, MA, and ARMA.

In the given context, which of the following statements are correct?

1. Even after deseasonalisation and trend estimation, the remaining deviations reflected in "percentage of trend" represent a combination of cyclical and irregular variations.
2. Since irregular variation cannot be mathematically isolated, it cannot be incorporated into any forecasting model, including AR and MA models.
3. A quarter with a "percentage of trend" significantly exceeding 100 necessarily indicates that seasonal adjustment has been incorrectly performed.
4. The ARMA model incorporates both lagged values and past error terms, enabling it to partially capture irregular fluctuations arising from shocks.
5. Irregular variations, being random and short-term, tend to neutralize over time and thus do not significantly distort long-term trend estimation.

A. 2, 3 and 4 only

B. 1, 2 and 3 only

C. 1, 4 and 5 only

D. 1, 3, 4 and 5 only

Answer: C

Solution:

Statement 1 – Correct: Once trend and seasonal components are removed, the remaining variation includes cyclical movements and irregular (random) fluctuations. These jointly influence the “percentage of trend.”

Statement 2 – Incorrect: Even though irregular variation cannot be isolated precisely, AR and MA models account for randomness through stochastic error terms. The MA component, in particular, models past shocks, thereby incorporating irregular effects indirectly.

Statement 3 – Incorrect: A value above 100 simply indicates that actual performance exceeds the trend. Since seasonal effects are already removed, such deviations are more likely due to cyclical expansion or irregular shocks rather than faulty seasonal adjustment.

Statement 4 – Correct: ARMA combines:

Autoregression (past values)

Moving average (past error terms)

This dual structure allows the model to statistically capture irregular fluctuations caused by unforeseen events.

Statement 5 – Correct: Irregular variations are non-systematic and short-lived. Over time, their effects tend to offset each other, ensuring that long-term trend estimation remains largely unaffected.

6. Which of the following statements correctly describe aspects of Time Series analysis and its variations?

1. **Secular trend** refers to long-term random fluctuations in data that occur without any observable direction.

2. Cyclical fluctuations are characterized by periodic changes in economic activities that follow a regular, fixed interval pattern.

3. Seasonal variations involve regular changes that repeat over a year due to climatic, social, or economic causes.

4. Irregular variations are unpredictable and typically caused by unforeseen events like natural disasters or wars.

5. A combination of different time series components can often be observed in real-world data, not just one type of variation.

A) 1, 2, 3
B) 1, 3, 4, 5
C) 2, 3, 4
D) 3, 4, 5

Answer: D

Explanation:

- **Statement 1 - Incorrect:** A **secular trend** is **not random**; it represents a **long-term, smooth, directional movement** in data (e.g., steady increase in prices). Randomness is associated with irregular variation, not trends.

- **Statement 2 - Incorrect: Cyclical fluctuations** do **not follow fixed intervals** or regular patterns. Business cycles, for instance, vary in **length and amplitude**, making them irregular in nature despite being cyclical.

- **Statement 3 - Correct: Seasonal variation** represents **predictable, recurring changes** within a year, often due to **weather, festivals, or social behavior**, like increased flu in winter or tourism in summer.
- **Statement 4 - Correct: Irregular variations** are caused by **unexpected, non-recurring events**, such as **wars, floods, or earthquakes**, which make them **non-measurable** through standard time series models.
- **Statement 5 - Correct:** In reality, a time series **usually includes a mix** of components (trend, seasonal, cyclical, irregular), and effective analysis must **decompose** and study each component properly.

7. Which of the following statements related to **Trend Analysis in Time Series** are correct?

1. A secular trend refers to the short-term fluctuations in a time series due to seasonal or irregular variations.
2. The equation y^A=a+bx is used to represent a linear trend, where yA is the dependent variable and x is typically time.
3. Curvilinear or parabolic trends are better suited when a straight-line model does not sufficiently represent the data.
4. Studying past unusual events such as wars can help project future trends if a similar event occurs again.
5. When using second-degree trends, the relationship between time and the variable is assumed to be exponential in nature.

A) 1, 2, 3

B) 1, 3, 4, 5

C) 2, 3, 4

D) 3, 4, 5

Answer: C

Explanation:

- **Statement 1: Incorrect:** A *secular trend* represents **long-term** directional movement in a time series, not short-term fluctuations. Shortterm variations typically fall under *seasonal* or *irregular* variations.

- **Statement 2: Correct** The equation $y^A=a+bx$ is a **linear trend** equation used in time series analysis. Here, a is the **intercept**, b is the **slope**, x is the **independent variable (usually time)**, and y^A is the **estimated dependent variable**.

- **Statement 3: Correct**

  When the data does not follow a straight-line trend well, **curvilinear or parabolic trends** are used. These can capture non-linear behavior over time better than linear models.

- **Statement 4: Correct**

  The analysis mentions that even a sudden change in the past due to a major event (like war) can help in predicting future changes if similar events occur again. This makes trend analysis a tool for learning from historical patterns.

  - **Statement 5: Incorrect**

    Second-degree trends involve a **parabolic** relationship (polynomial of degree 2), not an **exponential** one. An exponential trend would involve a variable raised to a power, but here the form is $y=a+bx+cx^2$ a polynomial.

8. Which of the following statements is/are correct regarding cyclical and seasonal variations in time series analysis?

1. Cyclical variations complete their cycle within a year and repeat in a fixed manner, making them similar to seasonal variations.

Season

2. Depersonalization is performed to isolate and study components such as trend, cyclical, and irregular variations more accurately.

3. The Ratio to Moving Average method is used to compute seasonal indices, which reflect variation around a base of 100.

4. Cyclical variations are computed using residual methods such as dividing actual data by predicted trend values.

5. Seasonal variations are irregular fluctuations that cannot be predicted and must be ignored in serious time series forecasting.

A) 1, 2, 3, 4

B) 1, 3, 4, 5

C) 2, 3, 4

D) 1, 2, 3, 4, 5

Answer: C

Explanation:

1. **Incorrect** - *Cyclical variations* extend **beyond a year** and do **not** follow a fixed or regular pattern within each year. In contrast, *seasonal variations* are **predictable and repeat annually**, such as quarterly sales or monthly temperature cycles.

2. **Correct** - *Depersonalization* is a technique used to remove the effect of seasonal variation from time series data. This helps analysts study underlying *trend, cyclical,* and *irregular components* more effectively.

3. **Correct** - The *Ratio to Moving Average* method is a standard approach to calculate *seasonal indices.* These indices are typically expressed with a base value of **100**, and deviations from this base reflect the degree of seasonal influence.

4. **Correct** - One method to estimate *cyclical variation* is to use the **residual method**, where we calculate the ratio of actual value (y) to predicted trend value (y), i.e., **(y / y) x 100**. This reveals the cyclical impact relative to the trend.

5. **Incorrect** - *Seasonal variations* are **not irregular**; they are **predictable and repetitive**, often tied to calendar events or weather seasons. They are crucial in time series analysis and must be accounted for, not ignored.

9. Which of the following statements correctly relate to irregular variation and forecasting techniques in time series analysis?

1. Irregular variations can be mathematically isolated and predicted using ARMA models with high accuracy.
2. An unusually cold winter leading to increased electricity consumption is an example of irregular variation.
3. The Autoregressive (AR) model forecasts future values using only past values of the variable being predicted.
4. The Moving Average (MA) model accounts for past and present error terms while modeling the variable.
5. ARMA models combine both the effects of lagged values and past error terms for time series forecasting.

A) 1, 2, 3, 4
B) 1, 3, 4, 5
C) 1, 2, 3, 5
D) 2, 3, 4, 5

Answer: D **Explanation:**

Statement 1 - Incorrect:

Irregular variations are **random and unpredictable**

and **cannot be reliably isolated mathematically**.
While models like ARMA are powerful, they cannot
capture or predict purely irregular events such as natural disasters or sudden wars.
Statement 2 - Correct:
This is a classic example of irregular variation. An unusually cold winter leading to unexpected spikes in electricity usage is not part of any predictable seasonal, trend, or cyclical pattern.

Statement 3 - Correct:
The **Autoregressive (AR) model** works by using a **linear combination of past values** of the variable to forecast future values. It "regresses" the variable on itself over prior time lags.

Statement 4 - Correct:
The **Moving Average (MA) model** considers the **error terms** from prior time points to model the current value. This includes a combination of **contemporaneous and past errors**.

Statement 5 - Correct:
The **ARMA model** combines both the **autoregressive (AR)** and **moving average (MA)** parts, incorporating **past values and past residuals**

to generate predictions.

JAIIB CAIIB PREP

UNIT 6

Theory of Probability

STRUCTURE

Keywords

Conditional
one has already occurred

Second event occur

$\boxed{O + N = 1}$

$P = \dfrac{F}{\text{Total Event}}$

P, E, C → Max combination / form factorial sample

Max ways

Addition OR

Multiply And

UNIT 6 — PROBABILITY

- **Meaning**
- Chance of occurrence of an event.
- Probability always lies between:
- $0 \leq P(E) \leq 1$

Important Terms

Term	Meaning
Experiment	Activity performed
Event	Outcome
Sample Space	All outcomes

Important Formulas

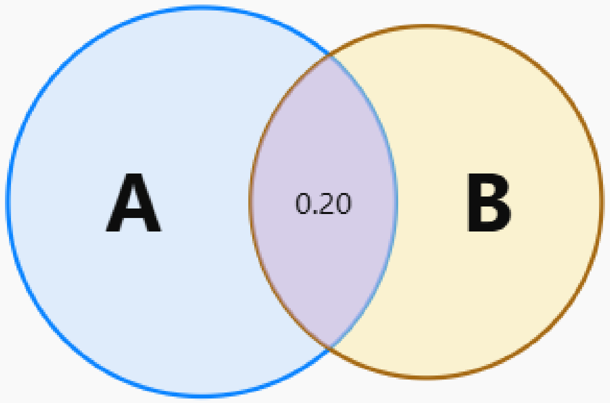
Addition Theorem

×

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$P(A)$	0.55	
$P(B)$	0.45	
$P(A \cap B)$	0.20	

$P(A \cup B) = P(A) + P(B) - P(A \cap B) \approx 0.80$



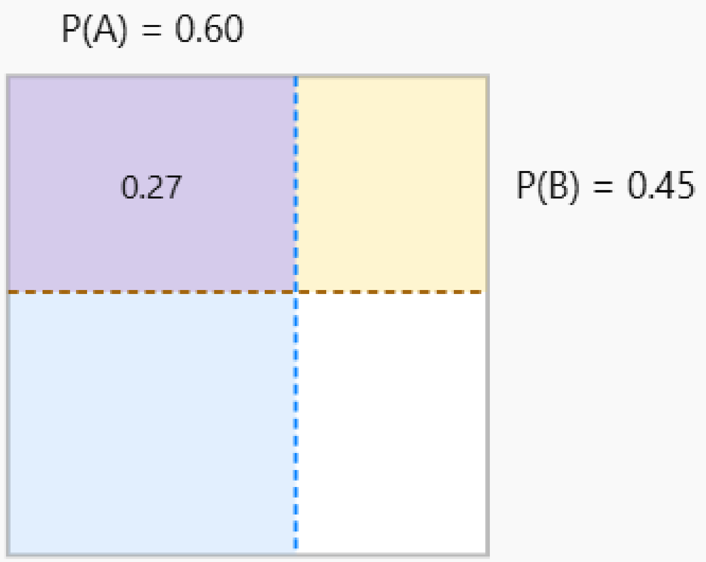
Multiplication Theorem

$$P(A \cap B) = P(A)P(B)$$

$P(A)$ 0.60

$P(B)$ 0.45

$P(A \cap B) = P(A) \cdot P(B) \approx 0.27$



Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

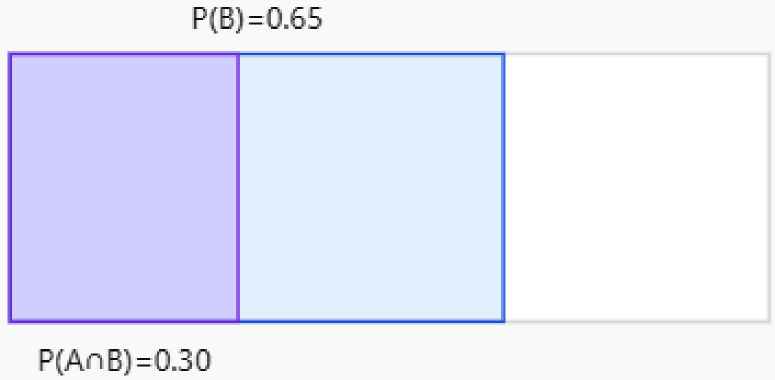
$P(B)$ 0.65

$P(A \cap B)$ 0.30

$P(A \mid B) = \frac{P(A \cap B)}{P(B)} \approx 0.46$

P(A|B) ≈ 0.46

A∩B is the part of B where A also happens



Bayes Theorem (VERY IMPORTANT)

Used in:

- Credit risk
- Fraud detection
- Decision-making

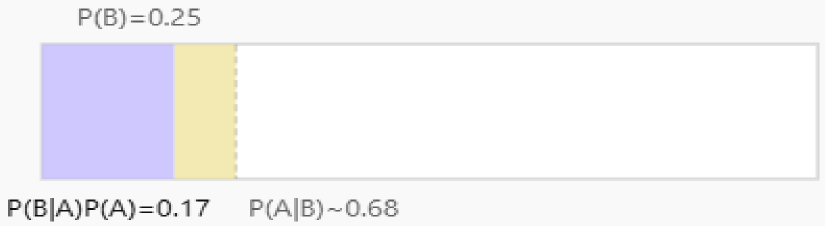
Formula:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

$P(A)$	0.20	
$P(B \mid A)$	0.85	
$P(B \mid \neg A)$	0.10	

$P(A \mid B) = \frac{P(B|A)P(A)}{P(B)} \approx 0.68,\ P(B) \approx 0.25$

Posterior = useful evidence / total evidence



PROBABILITY DISTRIBUTION

Meaning

Shows probability for different values of random variable.

Types of Random Variables

Type	Example
Discrete	No. of defaults
Continuous	Interest rate

BINOMIAL DISTRIBUTION

Used when:

- only 2 outcomes,
- success/failure,
- fixed trials.

Example:

Loan repayment default or not.

Formula:

$$P(X = x) = \binom{n}{x} p^x q^{n-x}$$

POISSON DISTRIBUTION

Used for rare events.

Examples:

- Fraud cases
- Server failure

NORMAL DISTRIBUTION (MOST IMPORTANT)

Characteristics:

- Bell-shaped
- Symmetrical
- Mean = Median = Mode

Z-score formula:

$$Z = \frac{X - \mu}{\sigma}$$

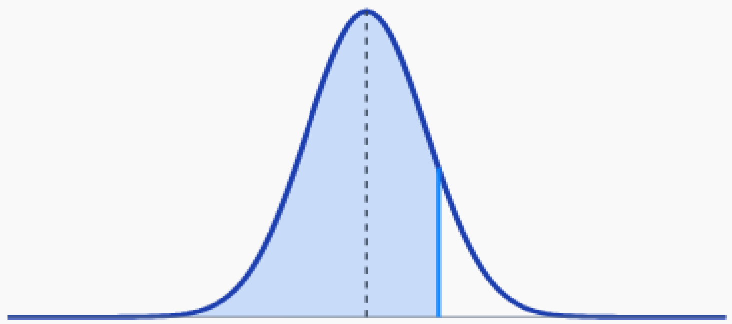
x 1.2

μ 0.0

σ 1.0

$z = \frac{x-\mu}{\sigma} \approx 1.2$

$\Phi(z) \approx 88.5\%$



1. A financial analytics firm is designing a risk simulation model based on random experiments involving selection and arrangement of analysts for a project.

R R R R R

- There are 8 analysts, out of which 5 are risk specialists and 3 are compliance specialists.
- The firm wants to:

1) R R C C
2) R R R C
3) R R R R

$^5C_2 \times {}^3C_2$ + $^5C_3\,{}^3C_1$ + $^5C_4\,{}^3C_0$

  1. Select a team of 4 analysts such that at least 2 are risk specialists.
  2. From the selected team, assign distinct roles (Leader, Reviewer, Auditor, Reporter) where order matters.
  3. Additionally, one analyst (from the remaining pool) is randomly selected as a backup, forming a random experiment.
  4. The probability that the backup analyst is a compliance specialist is to be evaluated.

Based on the above information, evaluate the following statements:

1. The total number of ways to select a team of 4 analysts with at least 2 risk specialists is equal to: $^5C_2 \cdot {}^3C_2 + {}^5C_3 \cdot {}^3C_1 + {}^5C_4 \cdot {}^3C_0$

$4! = 4 \times 3 \times 2 \times 1 = 24$

2. The total number of ways to assign distinct roles to the selected 4 analysts is equal to 4!4!4!, and this is independent of the method of selection.
3. The total number of sample points in the experiment of selecting one backup analyst from the remaining pool is 3.
4. The probability that the backup analyst is a compliance specialist is 34\frac{3}{4}43, irrespective of the selection of the main team.
5. The relation P(n,r)=r!·C(n,r) implies that total role assignments can also be computed as P(4,4).

A. Only 1, 2 and 5 are correct
B. Only 1, 2, 3 and 5 are correct
C. Only 2, 3 and 4 are correct
D. Only 1, 2, 4 and 5 are correct

Answer: A

Solution:

Statement 1: Correct: “At least 2 risk specialists” means possible cases: 2 risk, 2 compliance →$^5C_2 \cdot {}^3C_2$; 3 risk, 1 compliance → $^5C_3 \cdot {}^3C_1$; 4 risk, 0 compliance → $^5C_4 \cdot {}^3C_0$

Statement 2: Correct: Once 4 analysts are selected, assigning distinct roles is a permutation problem: 4!=24This does not depend on how the team was selected, only on the number of selected individuals.

Statement 3: Incorrect : Total analysts = 8; Selected = 4; Remaining = 4. So sample space size = 4.

Statement 4: Incorrect: Probability depends on how many compliance specialists remain after selection, which varies:

Example cases:

If 2 compliance selected → remaining compliance = 1 → probability = 1/4

If 1 compliance selected → remaining compliance = 2 → probability = 2/4

If 0 compliance selected → remaining compliance = 3 → probability = ¾

Statement 5: Correct: Given relation: $P(n,r)=r!\cdot C(n,r)$

For $n=4, r=4$: $P(4,4)=4!\cdot C(4,4)=4!\cdot 1=24$

2. A risk analytics team in a bank is designing a probabilistic model for fraud detection using a controlled experiment. The experiment involves rolling a fair die once and simultaneously drawing one card from a set of cards numbered $\{1, 2, 3, 4, 5, 6\}$.

The sample space is defined as:

$S = \{(i, j) \mid i = \text{outcome of die}, j = \text{outcome of card draw}, i, j \in \{1,2,3,4,5,6\}\}$

The following events are defined:

- A: Outcome where the die shows an odd number and the card drawn is even
- B: Outcome where the sum of die and card is divisible by 3
- C: Outcome where both die and card show the same number
- D: Outcome where the die shows a prime number
- E: Outcome where the card drawn is greater than 4

(3,3) = 6

P(O) = 1,3,5

ϕ

P(E) = 2,4,6

ϕ

D = 2 C = 2

Consider the following statements:

1. Events A and C are mutually exclusive.
2. Events B and C are not mutually exclusive.
3. Events D and E are independent but not mutually exclusive.
4. Events A and (A ∪ B) are exhaustive but not complementary.
5. The complement of event D consists of outcomes where the die shows $\{1, 4, 6\}$.

(A) Only 1, 2 and 5 are correct
(B) Only 2, 3 and 4 are correct
(C) Only 1, 2, 3 and 5 are correct
(D) Only 1, 3, 4 and 5 are correct

Answer: C

Solution:

Statement 1: True: A requires (odd, even) → pairs like (1,2), (3,4), etc. C requires ($i = j$) → (1,1), (2,2), etc. No overlap possible → Mutually exclusive

Statement 2: True: Example: (3,3) → sum = 6 (divisible by 3) and also same number. So $B \cap C \neq \emptyset$ → Not mutually exclusive

Statement 3: True: D = die shows prime → {2,3,5}; E = card > 4 → {5,6}. They relate to different components (die vs card) → Independent. But can occur together → e.g., (2,5) → Not mutually exclusive

Statement 4: False: $A \cup B$ already includes A → union cannot cover entire sample space S. Hence A and ($A \cup B$) are not exhaustive. Therefore statement is incorrect

Statement 5: True: D = {2,3,5} (prime outcomes of die), Complement = {1,4,6}

3. A financial analytics firm designs a probabilistic risk engine based on classical probability assumptions (equally likely, mutually exclusive outcomes). The system performs three independent experiments simultaneously:

1. Experiment 1: Two unbiased dice are thrown. Let events: A: Sum is 8; B: Both dice show same number
2. Experiment 2: Two balls are drawn simultaneously from a bag containing 7 white and 8 black balls. Let event: C: At least one white ball is drawn
3. Experiment 3: All letters of the word “MARKET” are randomly arranged.
   Let events: D: Vowels occupy both extreme positions; E: Vowels and consonants alternate starting with a consonant

Further, define event F = (A ∪ B) and event G = (C ∩ D).

Which of the following statements is correct?

1. $P(A)=5/36$ and $P(B)=1/6$, hence $P(F)=11/3$
2. Events A and B are mutually exclusive, hence $P(A \cap B)=0$.
3. $P(C)=1-8C2/15C2$
4. $P(D)=2!\cdot 4! / 6!$
5. Events C and D are independent, hence $P(G)=P(C)\cdot P(D)$.
6. $P(E)=3!\cdot 3! / 6!$
7. $P(C^c)=8C2/ 15C2$ and hence $P(C)+P(C^c)=1$

A. Only 1, 3, 4, 6 and 7 are correct
B. Only 1, 3, 5 and 7 are correct
C. Only 2, 4, 5 and 6 are correct
D. Only 3, 4 and 7 are correct ✓

or

And

W W
W B
B B

D_1 6 × 6 D_2 = 36

(2,6) (6,2) $\boxed{4,4}$ (3,5) (5,3)

$\frac{5}{36}$

B
1,1
2,2
$\boxed{4,4}$
3,3
5,5
6,6

$\frac{6}{36} = \frac{1}{6}$

P(A ∪ B)
Addition Theorem

$\frac{5}{36} + \frac{1}{6} - \frac{1}{36} = \frac{10}{36}$

Answer: D

Solution:

Statement 1: Incorrect: $P(A)=5/36$, $P(B)=1/6=6/36$. But A and B overlap at (4,4): ⇒ $P(A\cap B)=1/36$ So: $P(A\cup B)=5/36+6/36-1/36=10/36$.

Statement 2: Incorrect: A (sum = 8) includes (4,4); B (same numbers) includes (4,4) ⇒ Not mutually exclusive ⇒ $P(A\cap B)\neq 0$

Statement 3: Correct: Total ways = 15C2; No white = both black = 8C2; $P(C)=1-(8C2/15C2)$

Statement 4: Correct: Word = MARKET (6 letters, vowels = A, E → 2 vowels): Extremes must be vowels → arranged in 2! Ways; Remaining 4 letters → 4! Ways $P(D)=2!\cdot 4!\ /\ 6!$

Statement 5: Incorrect: C → ball experiment; D → arrangement problem. They belong to different sample spaces → independence cannot be assumed directly in classical sense.

Statement 6: Incorrect: MARKET: Vowels = 2 (A, E); Consonants = 4. Alternating pattern starting with consonant requires equal vowels & consonants → not possible

Statement 7: Correct: $P(C^c)=8C2\ /\ 15C2$; Complement rule: $P(C)+P(C^c)=1$

4. A risk analytics team at a bank is studying the behavior of two independent market signals:

- Event A: A stock price increases in a given week
- Event B: Trading volume increases in the same week

From historical data:

- $P(A)=0.3$
- $P(B)=0.5$
- However, during volatile periods, dependence arises such that:
  - $P(A\cap B)=0.24$
- Additionally, a third event C (market sentiment positive) is introduced such that:
  - $P(C|A\cap B)=0.75$
  - $P(C|A\cap Bc)=0.4$
  - $P(C|Ac\cap B)=0.6$
  - $P(C|Ac\cap Bc)=0.2$

$P(A)+P(B)-P(A\cap B)$

Which of the following statements is correct?

1. Events A and B are independent under volatile conditions.
2. The probability that at least one of A or B occurs is 0.56.
3. The probability that neither A nor B occurs is 0.44.
4. The unconditional probability of event C is 0.494.
5. The conditional probability $P(A|B)$ is greater than $P(A)$.

A. Only 2, 4 and 5 are correct
B. Only 1, 3 and 4 are correct
C. Only 2, 3,and 5 are correct
D. Only 2, 4 and 5 are correct

Answer: C

Solution:

Statement 1: Incorrect: For independence: $P(A\cap B)=P(A)\cdot P(B)=0.3\times 0.5=0.15$. But given: $P(A\cap B)=0.24$. Hence events are not independent

Statement 2: Correct: $P(A\cup B)=P(A)+P(B)-P(A\cap B)$ $=0.3+0.5-0.24=0.56$

Statement 3: Correct: $P(A^c\cap B^c)=1-P(A\cup B)=1-0.56=0.44$

Statement 4: Incorrect: Break into 4 mutually exclusive cases:

$P(C)=\sum P(C|\text{case})\cdot P(\text{case})$. Compute individual probabilities: $P(A\cap B)=0.24$; $P(A\cap Bc)=P(A)-P(A\cap B)=0.3-0.24=0.06$; $P(A^c\cap B)=P(B)-P(A\cap B)=0.5-0.24=0.26$; $P(A^c\cap B^c)=0.44$.
$=0.18+0.024+0.156+0.088=0.448$

Statement 5: Correct: $P(A|B)=P(A\cap B)\ /\ P(B)=0.24\ /\ 0.5=0.48$. Compare with $P(A)=0.3$. Since $0.48>0.3$, the statement is correct.

5. A financial analytics firm studies customer flow and portfolio returns simultaneously.

Experiment A: Number of customers arriving at a bank counter in a fixed 10-minute interval is denoted by random variable X. The variable X can take values 0, 1, 2, ... with associated probabilities f(x), satisfying properties of a probability mass function (PMF).

Experiment B: Return (in percentage) on a high-frequency trading portfolio in a day is denoted by random variable Y, modeled using a continuous probability density function g(y), defined over a bounded interval.

Cumulative distribution functions $F_X(x)$ and $F_Y(y)$ are defined for both variables.

Based on the above information, evaluate the following statements and identify the correct ones.

1. Since X is countable, the probability that X assumes a specific value and must always be strictly positive.
2. For the continuous random variable Y, even if g(y) is greater than zero at a point, the probability P(Y = y) is equal to zero.
3. If $F_X(x)$ is the cumulative distribution function of X, then it must be strictly increasing for all values in its domain.
4. The function g(y) must satisfy the condition that the integral of g(y) over the entire real line equals 1, even if the actual range of Y is bounded.
5. If f(x) is a valid probability mass function, then the sum of f(x) over all x equals 1, but it is not necessary that each f(x) is less than or equal to 1.

A. Only 1, 2 and 4 are correct
B. Only 2 and 4 are correct
C. Only 2, 4 and 5 are correct
D. Only 1, 3 and 5 are correct

Answer: B

Solution:

Statement 1: Incorrect: A discrete random variable may assign zero probability to some outcomes. Hence, it is not necessary that every possible value has strictly positive probability.

Statement 2: Correct: For a continuous random variable, probability at a specific point is always zero, regardless of the value of the density function at that point.

Statement 3: Incorrect: A cumulative distribution function is non-decreasing, but not necessarily strictly increasing. It can remain constant over intervals.

Statement 4: Correct: The total area under a probability density function over the entire real line must be equal to 1. Outside the bounded interval, the density is zero.

Statement 5: Incorrect: Each value of a probability mass function must lie between 0 and 1. Therefore, it is necessary that f(x) is less than or equal to 1 for all x.

6. A risk analyst models the return X of a structured derivative as a discrete random variable with the following probability distribution:

X: 0 1 2 3
f(x): k 2k 3k (1 − 6k)

The analyst defines a transformed variable: $Y = 3X - 2$

Further, another variable Z is defined as follows:
If $X \leq 1$, then $Z = X$
If $X > 1$, then $Z = 2X - 1$

Based on the above information, examine the following statements:

1. The value of k is 0.1.
2. The expected value E(X) is equal to 1.8.
3. The variance of Y is 9 times the variance of X.
4. The expected value E(Z) is greater than E(X).
5. The cumulative distribution function F(x) at x = 2 is 0.6.

A. Only 1, 3, 4 and 5 are correct
B. Only 1, 3 and 4 are correct
C. Only 2, 3 and 5 are correct
D. Only 1, 4 and 5 are correct

$Var(ax+b) = a^2 Var(x)$

$Var\,Y = 9\,Var(x)$

$k + 2k + 3k + (1-6k) = 1$

$6k + 1 - 6k = 1$

$1 - 6k \geq 0$

$k \leq \frac{1}{6}$

$0(k) + 1(2k) + 2(3k) + 3(1-6k) = 2$

$\boxed{k = .1}$

Answer: A

Solution:

Statement 1: Correct: Sum of probabilities must be equal to 1: $k + 2k + 3k + (1 - 6k) = 1$

$6k + 1 - 6k = 1$. Thus, equation holds for all k, but probability must be non-negative: $1 - 6k \geq 0 \Rightarrow k \leq 1/6$

Taking $k = 0.1$ satisfies all conditions.

statement 2: Incorrect. Expected value E(X): $E(X) = 0k + 1(2k) + 2*(3k) + 3*(1 - 6k) = 2k + 6k + 3 - 18k = 3 - 10k$. Putting $k = 0.1$: $E(X) = 3 - 1 = 2$

statement 3: Correct: For $Y = 3X - 2$: Variance property: $Var(aX + b) = a^2 Var(X)$ So, $Var(Y) = 9 Var(X)$

Statement 4: Correct: Compute E(Z): For $X = 0 \rightarrow Z = 0$; For $X = 1 \rightarrow Z = 1$; For $X = 2 \rightarrow Z = 3$; For $X = 3 \rightarrow Z = 5$

$E(Z) = 0k + 1(2k) + 3*(3k) + 5*(1 - 6k) = 2k + 9k + 5 - 30k = 5 - 19k$. Putting $k = 0.1$: $E(Z) = 5 - 1.9 = 3.1$

Since $E(X) = 2$ and $E(Z) = 3.1$, $E(Z) > E(X)$

Statement 5: Correct: CDF at $x = 2$: $F(2) = P(X \leq 2) = k + 2k + 3k = 6k$

Putting $k = 0.1$: $F(2) = 0.6$

7. A pharmaceutical company conducts a clinical trial where a new drug is tested on 20 patients. Each patient independently results in either success (probability p) or failure. Let X denote the number of successes.

The following observations are made:

- The distribution of X is negatively skewed
- The variance of X is 3.2
- The distribution has two modes
- The mean number of successes exceeds 12
- The kurtosis is less than 3

Based on the above, evaluate the following statements:

1. The value of p must be greater than 0.5.
2. The value of (n + 1)p is an integer.
3. The value of p is 0.6.
4. The condition for bimodality implies that np must be an integer.
5. The given variance implies that p(1 - p) = 0.16.

Which of the following is correct?

A. 1, 2, 3 and 5 only
B. 1, 2 and 5 only
C. 1, 2, 4 and 5 only
D. 2, 3 and 4 only

$p > .5$

$p = .8$ @ $.2$

$V = np(1-p) = 3.2$

$n = 20$

$20p(1-p) = 3.2$

$p(1-p) = \frac{3.2}{20} = .16$

$p(1-p) = .16$

$p - p^2 = .16$

$p^2 - p - .16$

Answer: B

Solution:

Given n = 20 and variance = np(1 - p) = 3.2

So, 20 × p(1 - p) = 3.2
Therefore, p(1 - p) = 0.16

Solving:
$p^2 - p + 0.16 = 0$
p = 0.8 or 0.2

Since distribution is negatively skewed, $p > 0.5$, so p = 0.8

Statement 1: Correct. Negative skewness implies $p > 0.5$.
Statement 2: Correct. Bimodality requires (n + 1)p to be an integer (as per condition given).
Statement 3: Incorrect. p is solved as 0.8.
Statement 4: Incorrect. Condition applies to (n + 1)p, not np.
Statement 5: Correct. Derived directly from variance.

8. A telecom company analyzes two independent random phenomena:

- The number of wrong calls received per hour follows a Poisson Distribution with mean 4.
- The duration (in minutes) of calls follows a Normal Distribution with mean 10 minutes and standard deviation 2 minutes.

Further, the company observes that:

- On a randomly selected hour, the skewness and kurtosis of the count distribution are evaluated.
- A standardized variable Z is constructed for call duration.
- The management is interested in understanding modal behavior and dispersion properties.

Based on the above, examine the following statements:

1. The probability that exactly 3 wrong calls occur in an hour is given by $P(X=3) = \dfrac{e^{-4} \cdot 4^3}{3!}$

$$\frac{e^{-\lambda}\lambda^{x}}{x!} = \frac{e^{-4}4^{3}}{3!}$$

2. The coefficient of skewness for the number of wrong calls is less than 1, indicating near symmetry.
3. If λ = 4 (an integer), the distribution of wrong calls is bimodal with modes at 3 and 4.
4. For the call duration distribution, after standardization, the transformed variable ZZZ will have mean 0 and variance 1.
5. The quartile deviation of call duration is equal to 0.6745 * 2.

A. 1, 3, 4 and 5 only

B. 1, 2, 3 and 4 only

C. 1, 4 and 5 only

D. 2, 3, 4 and 5 only

$.6745\sigma$

$\frac{X-\mu}{\sigma}$

Answer: A

Solution:

Statement 1:Correct: For Poisson distribution: $P(X=x)=\frac{e^{-\lambda}\lambda^{x}}{x!}$. Substituting λ = 4 and x = 3 gives the correct expression.

Statement 2: Incorrect: Skewness of Poisson distribution = $\frac{1}{\sqrt{\lambda}}$ For λ = 4 → skewness = 1/2 = 0.5. Although it is less than 1, Poisson distribution is still positively skewed, not symmetric.

Statement 3: Correct: When λ is an integer, Poisson distribution is bimodal with modes at: λ and λ − 1 → 4 and 3.

Statement 4: Correct: Standardization: $Z=\frac{X-\mu}{\sigma}$ Always results in mean = 0 and variance = 1.

Statement 5: Correct: Quartile deviation of normal distribution = 0.6745σ

Here σ = 2; QD = 0.6745 × 2.

9. A large commercial bank headquartered in India has extended the following credit exposures:

- ₹120 crore term loan to Company A (steel sector), with high dependence on global commodity prices.
- ₹80 crore exposure to Company B (same steel sector), showing early stress signals and 75 days overdue.
- ₹150 crore sovereign bonds issued by Country X, where recent political instability has raised concerns about repayment capacity.
- ₹50 crore retail loan portfolio, where historical Probability of Default (PD) is 4%, Exposure at Default (EAD) is full, and estimated Loss Given Default (LGD) is 60%.

The bank uses the expected loss model to estimate credit risk and follows a 90-day overdue norm for default classification.

1. If Company B crosses 90 days overdue, the exposure will be categorized under credit default risk irrespective of sector concentration.
2. The bank's combined exposure to Company A and Company B represents concentration risk even if both borrowers are currently performing.
3. Investment in sovereign bonds of Country X exposes the bank to country risk only if the country has already defaulted on its obligations.
4. Expected loss for the retail portfolio should be computed as: EL = PD × EAD × (1 – LGD), implying that higher LGD reduces expected loss.
5. If PD = 4%, EAD = ₹50 crore, and LGD = 60%, the expected loss on the retail portfolio will be ₹0.8 crore.

A. 2, 3 and 5 only
B. 1, 2 and 4 only
C. 1, 2 and 5 only
D. 1, 2, 3 and 5 only

Answer: C

Solution:

Statement 1 – Correct: Once the overdue period crosses 90 days, it qualifies as default. This falls under credit default risk, regardless of whether the exposure is concentrated in a sector or not.

Statement 2 – Correct: Both exposures are in the same steel sector, creating vulnerability to sector-specific shocks. Even if both are performing, the bank faces concentration risk due to lack of diversification.

Statement 3 – Incorrect: Country risk arises from the possibility that a sovereign may be unwilling or unable to pay—not only after actual default. Political instability itself triggers country risk.

Statement 4 – Incorrect: Formula given: $EL = PD \times EAD \times (1 - LGD)$

Here, $(1 - LGD)$ represents recovery rate. So, higher LGD → lower recovery → higher expected loss, not lower.

Statement 5 – Correct: $EL = 4\% \times 50 \text{ crore} \times (1 - 0.60) = 0.04 \times 50 \times 0.40 = 0.8$ crore

10. A portfolio manager at a large investment firm manages a diversified portfolio consisting of equity, gold ETFs, and debt mutual funds. The current market value of the portfolio is ₹50 crore. Based on historical analysis, the expected annual return of the portfolio is 12% and the standard deviation (volatility) is 18%.

The risk committee asks the manager to compute and interpret the Value at Risk (VaR) at both 95% and 99% confidence levels for a 1-year horizon, assuming normal distribution of returns.

Further, the committee raises concerns regarding the adequacy and limitations of VaR as a risk measure, especially in stressed market conditions.

Based on the above information, which of the following statements is correct?

$[\text{Return} - 1.65 \times \sigma] \times 50\text{cr}$

$= [.12 - (1.65 \times .18)] 50$

1. The 1-year VaR at 95% confidence level for the portfolio is approximately ₹8.85 crore.
2. The 1-year VaR at 99% confidence level will necessarily be lower than the VaR at 95% confidence level due to higher confidence.
3. VaR captures only downside risk and ignores potential gains, thereby addressing a key limitation of volatility as a risk measure.
4. The calculation of VaR in this case relies on the assumption that returns are normally distributed and volatility remains constant.
5. VaR does not provide any information about losses beyond the specified confidence level, making it insufficient in extreme market scenarios.

(A) 1, 3, 4 and 5 are correct

(B) 1, 2, 3 and 4 are correct

(C) 2, 3, 4 and 5 are correct

(D) 1, 3 and 5 are correct

Answer: A

Solution:

Statement 1: Correct: VaR (95%) = (Return−1.65×σ)×Portfolio Value = (0.12−1.65×0.18)×50 crore = 0.177×50 ≈ −₹8.85 crore. This represents the maximum expected loss at 95% confidence.

Statement 2: Incorrect: At higher confidence (99%), the multiplier (2.33) is higher: VaR becomes more conservative (larger loss), not lower. Hence, VaR at 99% > VaR at 95%.

Statement 3: Correct: Unlike volatility, which measures both upside and downside, VaR focuses only on downside risk (loss probability).

Statement 4: Correct: The formula used:

- Assumes normal distribution of returns
- Uses standard deviation as constant volatility

These are strong assumptions and limitations

Statement 5: Correct: VaR answers: “What is the maximum loss at X% confidence?” But it does NOT answer: “What happens beyond that loss?” Thus, it ignores tail risk / extreme losses (black swan events).

11. A portfolio manager is evaluating hedging strategies using derivative instruments on a stock currently trading at ₹100. The following market parameters are observed:

- Strike Price (SP): ₹105
- Risk-free rate (r): 8% p.a.
- Time to maturity: 1 year
- Volatility: High and increasing
- Market expectation: Price may move sharply in either direction

The manager considers two alternatives:

- Strategy A: Enter into a forward contract to buy the stock at ₹105 after 1 year
- Strategy B: Buy a European call option with strike ₹105

At maturity, the stock price turns out to be ₹90.

Which of the following statements are correct?

1. Under Strategy A, the manager incurs a loss due to binding obligation, whereas under Strategy B, the loss is limited to the premium paid.
2. The opportunity loss exists in Strategy A only when price falls below strike price, but not when price rises above strike price.
3. Increase in volatility prior to maturity would have increased the value of Strategy B but would have no effect on Strategy A.
4. If interest rates had increased significantly during the contract period, the value of the call option would increase while the forward price would also increase.
5. Even if the option expires worthless, its initial valuation must have incorporated both intrinsic value and time value components.

Which of the following is correct?

A. 2, 4 and 5 only
B. 1, 2, 3, 4 and 5
C. 1, 3 and 4 only
D. 1, 3, 4 and 5 only

Answer: D

Solution:

Statement 1 – Correct: Forward: purchase at ₹105 → market price ₹90 → loss = ₹15; Call: option not exercised → loss limited to premium This is the core advantage of options (asymmetric payoff).

Statement 2 – Incorrect: Opportunity loss exists both ways:

- If price rises → forward is beneficial but still no flexibility
- If price falls → forced loss

Statement 3 – Correct: Volatility ↑ → option value ↑ (more uncertainty → more upside potential). Forward value not impacted by volatility (fixed contract)

Statement 4 – Correct: Call option: ↑ r → ↓ PV(SP) → call value increases; Forward price formula: F=S(1+r)t → ↑ r → forward price increases

Statement 5 – Correct: Even if final intrinsic value = 0, initially:

- Option price = intrinsic value + time value
- At initiation: intrinsic = 0 (OTM), but time value > 0

12. From a group of **6 men and 5 women**, a committee of **4 people** is to be formed. What is the number of ways in which the committee can be formed such that **at least 2 women** are included?

A. 210

B. 215

C. 240

D. 255

$${}^6C_0 \times {}^5C_4 + {}^5C_2 \times {}^6C_2 + {}^5C_3 \times {}^6C_1$$

W W   W W

W W   M M

W W W   W M

Answer: B

We are forming a committee of **4 people** from **6 men and 5 women**.

We need to find the number of ways in which the committee includes **at least 2 women**.

So we consider these three valid cases:

- Case 1: 2 women and 2 men
- Case 2: 3 women and 1 man
- Case 3: 4 women and 0 men

Case 1: 2 women and 2 men

- Ways to choose 2 women from 5 = $5C_2$ = 5x4/ (2x1) =10
- Ways to choose 2 men from 6 = 6C2 = 6x5/ (2xl)=l5
- Total = 10 x 15 = **150 Case 2: 3 women and 1 man**
- 5C3 = 5x4x3/ (3x2x1)=10
- 6C1 = 6
- Total = 10 x 6 = **60 Case 3: 4 women and 0 men**
- **5C4 = 5x4x3x2/ (4x3x2x1)=5**
- **6C0 = 1 (only one way to choose none)**
- **Total = 5 x 1 = 5**

Total number of ways = 150 + 60 + 5 = 215

13. Two unbiased dice are thrown simultaneously. Based on the experiment, which of the following statements is/are **correct**?

1. The probability that both dice show the same number is 1/6.
2. The probability that the first die shows a 6 is 5/36.
3. The probability that the sum of numbers on both dice is 8 is 5/36.
4. The probability that at least one die shows a 1 is 11/36.
5. The total number of possible outcomes when two dice are thrown is 30.

A. Only 1, 3 and 4
B. Only 1 and 3
C. Only 2, 4 and 5
D. Only 1, 3 and 5

Answer: A

Statement 1: The probability that both dice show the same number is 1/6.

Favourable outcomes: (1,1), (2,2), (3,3), (4,4), (5,5), (6,6) → 6 outcomes

Total outcomes: 6 × 6 = 36

Probability = 6/36 = **1/6**

Statement 2: The probability that the first die shows a 6 is 5/36.

First die fixed at 6, second die can be anything → (6,1), (6,2), (6,3), (6,4), (6,5), (6,6) → 6 outcomes

So, probability = 6/36 = **1/6**, not 5/36

Statement 3: The probability that the sum of numbers on both dice is 8 is 5/36.

Combinations: (2,6), (3,5), (4,4), (5,3), (6,2) → 5 favourable outcomes

Probability = 5/36

Statement 4: The probability that at least one die shows a 1.

Total outcomes = 36

Outcomes where **no die shows 1**:

Choices for each die: 2,3,4,5,6 → 5 options each → 5×5 = 25 outcomes

So, outcomes where at least one die shows 1 = 36 – 25 = 11

Probability = 11/36

Statement 5: The total number of possible outcomes when two dice are thrown is 30.

This is incorrect. Total outcomes = 6 × 6 = **36**, not 30

14. Which of the following is not a property of the Cumulative Distribution Function (CDF)?

A. F(x) is always a non-decreasing function

B. F(-^) = 0 and F(+^) = 1

C. F(x) is continuous from the left

D. F(x) lies between 0 and 1 for all x

Answer: C

The CDF, F(x) = P(X < x), has the following properties:

- Non-decreasing function
- F(-^) = 0, F(+^) = 1
- Right-continuous, not left-continuous
- Takes values between 0 and 1

JAIIB CAIIB PREP

UNIT 7 Estimation

STRUCTURE

Back at 10:35

UNIT 7 — ESTIMATION

- **1. Meaning of Estimation**
- Estimation means:
  Using sample data to estimate population values.
- Example:
- Estimating average bank deposits
- Estimating customer income
- Estimating NPA percentage
- Population value is unknown, so sample is used.

2. Types of Estimation

Type	Meaning
Point Estimation	Single value estimate
Interval Estimation	Range estimate

3. Point Estimation

Single value used to estimate population parameter.

Example:

Sample mean estimates population mean.

If sample average salary = ₹40,000

Then ₹40,000 is point estimate.

4. Interval Estimation

Gives a range within which parameter may lie.

Example:

Average income may lie between:

₹38,000 to ₹42,000

More reliable than point estimate.

5. Confidence Interval

Most important topic.

Confidence interval means:

Range within which population parameter lies with certain confidence.

Usually:

- 95% confidence
- 99% confidence

6. Confidence Level

Confidence level indicates reliability of estimate.

Example:

95% confidence means:

95 times out of 100 the estimate is correct.

7. Standard Error (Very Important)

- Measures sampling fluctuation.
- Formula:
- $SE = \frac{\sigma}{\sqrt{n}}$
- Important MCQ:
- Larger sample → lower SE
- Lower SE → better estimate

8. Estimation of Population Mean

When population SD known:

- $\bar{X} \pm Z\frac{\sigma}{\sqrt{n}}$

9. Estimation of Population Proportion

- Formula:
- $p \pm Z\sqrt{\frac{pq}{n}}$
- Where:
- p= proportion
- $q = 1 - p$

UNIT 8 — LINEAR PROGRAMMING

- **1. Meaning of Linear Programming (LP)**
- Linear Programming is a mathematical technique used to:
- maximize profit
- minimize cost
  subject to constraints.
- Very important for decision-making.

2. Uses in Banking

- Used for:
- Fund allocation
- Portfolio management
- Profit maximization
- Resource allocation

3. Components of LP

(A) Decision Variables

Unknown quantities.

Example:
Amount invested in loans and securities.

(B) Objective Function

Function to maximize/minimize.

Example:
Profit maximization.

General form:

$$Z = aX + bY$$

(C) Constraints

Restrictions or limitations.

Example:
Limited funds or manpower.

(D) Non-Negativity Restriction

Variables cannot be negative.

$$X \geq 0, \; Y \geq 0$$

---

4. Assumptions of Linear Programming

Remember:

PLCD

Letter	Meaning
P	Proportionality
L	Linearity
C	Certainty
D	Divisibility

5. Methods of Solving LP

Graphical Method

Used when:

- only two variables.

Most important for CAIIB numericals.

Simplex Method

Used for large problems.

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6. Feasible Region

Area satisfying all constraints.

Optimal solution lies within feasible region.

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7. Optimal Solution

Best possible solution:

- maximum profit
  or
- minimum cost

Occurs at corner points.

Very important MCQ.

8. Advantages of LP

- Scientific decision-making
- Better utilization of resources
- Profit maximization
- Cost reduction

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9. Limitations of LP

- Assumes linear relationship
- Difficult for complex problems
- Certainty assumption unrealistic

UNIT 9 — SIMULATION

- **1. Meaning of Simulation**
- Simulation means:
  Imitating real-life situations using models.
- Used when actual experimentation is difficult or costly.

2. Objective of Simulation

- To study system behavior under different conditions.
- Helps management in:
- forecasting
- risk analysis
- decision-making

3. Uses in Banking

- Simulation used for:
- Queue management
- Risk analysis
- Cash management
- Loan default prediction
- ATM cash planning

4. Characteristics of Simulation

- Based on probability
- Uses random numbers
- Represents real-world situations
- Helps evaluate alternatives

5. Types of Simulation

- **Monte Carlo Simulation (Most Important)**
- Uses:
- random numbers
- probability distributions
- Widely used in finance and banking.

6. Steps in Simulation

- Define problem
- Construct model
- Generate random numbers
- Perform experiment
- Analyze results

7. Random Numbers

- Random numbers are:
- unpredictable
- irregular
- used for simulation trials

8. Advantages of Simulation

- Handles complex problems
- Safer than real experimentation
- Cost-effective
- Useful for forecasting

---

9. Limitations of Simulation

- Time-consuming
- Requires expertise
- Results depend on assumptions
- Not exact solution

MOST IMPORTANT FORMULAS

Standard Error

$$SE = \frac{\sigma}{\sqrt{n}}$$

Confidence Interval

$$\bar{X} \pm Z\frac{\sigma}{\sqrt{n}}$$

LP Objective Function

$$Z = aX + bY$$

---

1. A large commercial bank is attempting to estimate the default probability (population proportion) of a new category of borrowers. Due to time constraints, the credit risk team selects a random sample of past borrowers and computes multiple statistics to infer about the population parameter.

The Head of Risk Analytics presents the following approaches:

- Approach A: Uses sample mean default rate as a point estimate.
- Approach B: Provides a range of default rates (interval estimate) with an associated confidence level.
- Approach C: Uses an alternative statistic with higher variability but simpler computation.
- Approach D: Recommends increasing sample size significantly to improve estimation reliability.
- Approach E: Suggests replacing the estimator with another statistic that extracts additional unused information from the same sample.

Based on the above scenario, which of the following statements are correct?

1. A point estimate provided in Approach A is inherently limited because it does not convey the magnitude of possible estimation error or reliability.
2. Approach B improves decision-making quality because an interval estimate communicates both the range of plausible values and the likelihood of containing the true population parameter.
3. If Approach C uses a statistic with a higher standard error than Approach A, then it is necessarily a more efficient estimator due to computational simplicity.
4. The recommendation in Approach D will improve estimation reliability only if the estimator used is consistent; otherwise, increasing sample size may not yield better convergence to the true parameter.
5. Approach E contradicts the property of sufficiency, because a sufficient estimator already utilizes all relevant sample information and cannot be improved upon using the same data.

(A) Only 1, 2 and 4 are correct
(B) Only 1, 2, 4 and 5 are correct
(C) Only 2, 3 and 5 are correct
D) Only 1, 3, 4 and 5 are correct

Answer: B

Solution:

Statement 1 – Correct: A point estimate is a single value. Without an accompanying measure of dispersion or confidence, it gives no indication of reliability or possible deviation from the true population parameter.

Statement 2 – Correct: An interval estimate provides:

- A range (lower and upper bounds)
- A probability (confidence level)

This dual information makes it far more useful for decision-making under uncertainty.

Statement 3 – Incorrect: Efficiency depends on standard error, not computational simplicity.

Higher standard error: less efficient estimator

Lower standard error: more efficient estimator

Hence, Approach C is inferior in efficiency.

Statement 4 – Correct: Consistency ensures that as sample size increases, the estimator converges toward the true population parameter. If estimator is not consistent, increasing sample size does not guarantee improved accuracy. Thus, blindly increasing sample size may waste resources.

Statement 5 – Correct: Sufficiency implies: The estimator uses all relevant information in the sample. No other estimator can extract additional useful information from the same data. Hence, proposing a better estimator from the same dataset contradicts sufficiency.

2. A large commercial bank is estimating key parameters for decision-making related to cash handling and operational risk. The bank randomly selects 35 branches and records the number of cash transactions processed per day in each branch. The sample mean is found to be 102 transactions per branch per day, with a sample standard deviation of 6.01.

In a separate operational risk assessment, the bank inspects 50 branches and finds that 4 branches reported cash handling discrepancies during the audit period.

Based on this information, evaluate the following statements and identify the correct ones.

1. The sample mean of 102 transactions is an unbiased, consistent, and efficient estimator of the population mean, and its sampling distribution can be approximated to normal due to the Central Limit Theorem.
2. If the bank uses n instead of n−1 in estimating variance, the resulting estimate will be biased downward and hence underestimate the true population variance.
3. The sample proportion of branches with discrepancies (0.08) serves as an unbiased and efficient estimator of the population proportion, and its precision improves as the sample size increases.
4. Increasing the number of sampled branches for estimating the mean will increase the standard error, since more variability across branches will be captured.
5. Since the estimators used are unbiased and efficient, the bank can rely on point estimates without any concern for sampling error in managerial decisions.

(A) Only 1, 2 and 3 are correct
(B) Only 2, 3 and 4 are correct
(C) Only 1, 2, 3 and 5 are correct
(D) Only 1, 3, 4 and 5 are correct

Answer: A

Solution:

Statement 1: Correct. The sample mean is a desirable estimator—it is unbiased, consistent, and efficient. With a sample size of 35, the Central Limit Theorem ensures that the sampling distribution of the mean approaches normality, enabling probabilistic inference.

Statement 2: Correct. Using denominator n ignores the loss of degrees of freedom in estimation, leading to systematic underestimation of variance. Hence, $n-1$ is used to obtain an unbiased estimate.

Statement 3: Correct. The sample proportion (0.08) has all desirable properties—unbiasedness, consistency, efficiency, and sufficiency. Larger samples reduce standard error, enhancing reliability.

Statement 4: Incorrect. Standard error of the mean is inversely related to sample size $(SE = \sigma/\sqrt{n})$

Increasing sample size reduces standard error, improving precision.

Statement 5: Incorrect. Even the best estimators are subject to sampling error. Point estimates simplify decision-making but do not eliminate uncertainty.

3. A pharmaceutical company conducts a statistical study to estimate the average shelf life of a newly developed drug. A random sample of 256 units is selected, and the sample mean shelf life is found to be 48 months. The population standard deviation is known to be 16 months.

The analytics team applies interval estimation techniques using the Central Limit Theorem and constructs multiple confidence intervals based on the standard error. They further evaluate how these intervals behave under repeated sampling and interpret their probabilistic significance.

In the given context, which of the following statements are $\boxed{\text{not correct?}}$

$S.E = \frac{\sigma}{\sqrt{n}} = \frac{16}{\sqrt{256}}$

1. The standard error of the sample mean is 1 month, and it measures the variability of sample means around the population mean rather than variability of individual observations.
2. The interval (48 ± 1) months represents a 55.3% confidence interval for the population mean under normal approximation.
3. If 2,000 independent samples of size 256 are drawn and 95.5% confidence intervals are constructed, exactly 1,910 of these intervals will contain the population mean.
4. Increasing the confidence level from 95.5% to 99.7% increases the width of the interval, while the standard error remains unchanged.
5. For a specific interval (46 to 50 months), it can be said that there is a 95.5% probability that the population mean lies within this interval.

(A) Only 1, 2 and 5
(B) Only 2, 3 and 4
(C) Only 2, 3 and 5
(D) Only 1, 3, 4 and 5

Answer: C

Solution:

Statement 1: Correct: Standard error = $\sigma / \sqrt{n} = 16 / \sqrt{256} = 16 / 16 = 1$ month. It reflects dispersion of sample means, not individual observations.

Statement 2: Incorrect: ±1 standard error corresponds to 68.3% coverage in a normal distribution

Statement 3: Incorrect: In repeated sampling, approximately 95.5% intervals will capture the true mean. But saying exactly 1,910 out of 2,000 ignores sampling variability.

Statement 4: Correct: Higher confidence → wider interval. Standard error depends only on σ and n, hence unchanged.

Statement 5: Incorrect: For a given interval, the population mean either lies within it or not. Confidence level refers to long-run frequency, not probability of a specific interval.

4. A market research firm in India conducts a study on the average monthly expenditure of urban households. Due to cost constraints, only one random sample of 400 households is selected. The sample mean expenditure is ₹18,000 $\bar{x}$ and the standard error of the mean is ₹1,000. The analyst constructs multiple confidence intervals using different confidence levels and interprets them for decision-making. Based on the above scenario, which of the following statements are correct?

S.E n

1. If a 90% confidence level is used, the confidence interval will be approximately ₹18,000 ± ₹1,640, and this interval will be narrower than the 95% confidence interval.
2. If a 99% confidence level is used, the interval will be wider because the corresponding standard error multiplier (≈ 2.58) increases the margin of error.
3. The statement “We are 95% confident that the population mean lies between two limits” implies that there is a 95% probability that the specific calculated interval contains the true population mean.
4. If repeated samples of the same size are drawn and intervals are constructed each time, approximately 95% of such intervals will contain the true population mean when using a 95% confidence level.
5. Reducing the width of the confidence interval necessarily increases the confidence level, making the estimate more precise and more reliable simultaneously.

A. Only 2, 3 and 4 are correct
B. Only 1, 3 and 5 are correct
C. Only 1, 2 and 4 are correct
D. Only 1, 2, 4 and 5 are correct

Answer: C

Solution:

Statement 1 – Correct: For 90% confidence, the z-value ≈ 1.64. Margin of error = 1.64 × 1000 = ₹1,640. Lower multiplier → narrower interval than 95% (z ≈ 1.96).

Statement 2 – Correct: For 99% confidence, z ≈ 2.58 → margin of error increases. Higher confidence → wider interval, reflecting more coverage but less precision.

Statement 3 – Incorrect: Confidence level does not mean probability that the specific interval contains the parameter. Instead, it refers to long-run frequency of intervals capturing the true mean.

Statement 4 – Correct: This is the correct interpretation of confidence intervals: If many samples are drawn, about 95% of constructed intervals will include the true mean.

Statement 5 – Incorrect:
Narrower interval → lower confidence
Higher confidence → wider interval
Thus, reducing interval width reduces confidence, not increases it.

5. A financial research firm is analyzing two independent datasets for policy modeling:

Scenario A (Large Sample, Known Population Variability):
The firm studies the average processing time (in minutes) of loan applications in a large banking network. The population standard deviation is known to be 15 minutes. A random sample of 225 applications shows a mean processing time of 120 minutes. A 98% confidence interval is to be constructed.

n

Scenario B (Finite Population, Unknown Variability):
A regulator studies the average monthly electricity consumption of households in a semi-urban region consisting $\boxed{\text{of } N = 1{,}200}$ households. A sample of n = 120 households gives a sample mean of 310 units and a sample standard deviation of 60 units. A 95% confidence interval is to be constructed. Since the sample exceeds 5% of the population, finite population correction is applicable.

$\frac{15}{\sqrt{225}}$

In the given context, which of the following statements are correct?

1. In Scenario A, the standard error of the mean is 1 minute, and the critical value used for a 98% confidence level will be approximately 2.33.
2. In Scenario A, if the confidence level were reduced to 90%, the confidence interval would become narrower due to a reduction in the critical value.
3. In Scenario B, the standard error calculated using finite population correction will be strictly less than $\boxed{60/\sqrt{120},}$ and the reduction depends on the sampling fraction.
4. In Scenario B, if the population size were extremely large (approaching infinity), the finite population correction factor would approach zero, significantly shrinking the standard error.
5. In Scenario B, despite using a large sample, replacing population standard deviation with sample standard deviation introduces estimation risk; however, due to large sample size, the normal distribution is still used as an approximation.

$\sqrt{\frac{N-n}{N-1}}$ σ 1

A. Only 1, 2, 3, and 5 are correct
B. Only 1, 3, 4, and 5 are correct
C. Only 2, 3, and 5 are correct
D. All statements are correct

Answer: A

Solution:

Statement 1: Correct: Standard error = $\sigma / \sqrt{n} = 15 / \sqrt{225} = 15 / 15 = 1$. For 98% confidence, $z \approx 2.33$

Statement 2: Correct Lower confidence level → smaller z-value → narrower interval

Statement 3: Correct: FPC factor = $\sqrt{[(N - n)/(N - 1)]} < 1$. Thus, adjusted SE is always less than $\sigma/\sqrt{n}$ (or $s/\sqrt{n}$)

Statement 4: Incorrect: As $N \to \infty$, FPC = $\sqrt{[(N - n)/(N - 1)]} \to 1$ (not zero) So standard error approaches $\sigma/\sqrt{n}$, not zero

Statement 5: Correct

- σ unknown → replaced with s
- This introduces estimation uncertainty
- But for large n, normal approximation is still valid

6. A large multinational corporation is assessing employee preference regarding opting for self-managed retirement plans instead of company-sponsored schemes. A statistical analyst draws a simple random sample of 200 employees and finds that 90 employees prefer self-managed retirement plans. The management wants to construct a 98% confidence interval for the true population proportion. The analyst decides to approximate the binomial distribution using a normal distribution.

n

Based on this information, which of the following statements are correct?

1. The condition for normal approximation is satisfied since both n·p and n·q exceed the minimum threshold required.
2. The estimated standard error of the proportion is approximately 0.035.
3. The critical value (Z-value) to be used for constructing the confidence interval at 98% confidence level is approximately 2.33.
4. The upper confidence limit will exceed 0.60.
5. If the sample size were reduced to 40 while keeping the same sample proportion, the width of the confidence interval would decrease.

A. 1, 2 and 3 only //
B. 2, 3 and 4 only
C. 1, 2, 3 and 4 only
D. 1, 3, 4 and 5 only

$n \times p = .45 \times 200 = 90$

$n \times q = .55 \times 200 = 110$

$\hat{p} = \frac{90}{200} = .45$

$q = 1 - .45 = .55$

$\boxed{\hat{p} \pm z \times S.E}$

0.6+

Answer: A

Solution:

Step 1: Compute sample proportion (p) and q
$p = 90 / 200 = 0.45$; $q = 1 - p = 0.55$

Step 2: Check condition for normal approximation
$n \times p = 200 \times 0.45 = 90$
$n \times q = 200 \times 0.55 = 110$
Both values are greater than 5, so normal approximation can be used.
Statement 1 is correct.

Step 3: Calculate standard error
Standard Error (SE) $= \sqrt{(p \times q / n)} = \sqrt{(0.45 \times 0.55 / 200)} = \sqrt{(0.2475 / 200)} = \sqrt{(0.0012375)} \approx 0.035$
Statement 2 is correct.

Step 4: Identify Z value for 98% confidence level: For 98% confidence level, $Z \approx 2.33$. Statement 3 is correct.

Step 5: Construct confidence interval: Confidence Interval $= p \pm Z \times SE = 0.45 \pm (2.33 \times 0.035) = 0.45 \pm 0.08155$

Upper limit $= 0.45 + 0.08155 \approx 0.532$; Lower limit $= 0.45 - 0.08155 \approx 0.368$. The upper limit does not exceed 0.60. Statement 4 is incorrect.

Step 6: Effect of reduction in sample size: Standard Error $= \sqrt{(p \times q / n)}$. If n decreases, SE increases, leading to a wider confidence interval. Hence, reducing sample size increases the width of the interval. Statement 5 is incorrect.

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UNIT
8

Linear Programming

STRUCTURE

8.0 Objectives

8.1 Introduction

8.2 Graphic Approach

8.3 Simplex Method

Keywords

Questions

Objective

1) Max Profit

2) Min cost

Resources limited

Man power

Resource

Machine

higher level

1. A firm produces two products: Product P (x) and Product Q (y). The firm aims to maximize its total contribution under multiple operational and financial constraints.

- Contribution per unit:
  1. Product P = ₹2,400
  2. Product Q = ₹1,800
- Constraints:
  1. Processing time: $4x + 3y \leq 24$ 1
  2. Finishing time: $2x + y \leq 10$
  3. Budget constraint (cash outflow restriction): $3x + 2y \leq 18$ 2
  4. Demand restriction: $x \leq 5$
  5. Non-negativity: $x, y \geq 0$

Additional insights:

- At optimality, processing time and budget constraints are binding.
- Finishing time has unused capacity at the optimal point.
- The slope of the objective function is flatter than the processing constraint but steeper than the budget constraint.
- Fractional production is allowed.

Which of the following statements are correct?

1. The optimal solution lies at the intersection of processing time and budget constraints.
2. Finishing time constraint is non-binding at optimality and hence does not influence the optimal solution.
3. If the demand restriction on Product P is removed, the optimal solution will definitely shift to a new point.
4. Imposing integer constraints may reduce the maximum achievable contribution.
5. If the budget limit is increased while keeping other constraints unchanged, the optimal contribution will necessarily increase.

A. 1, 2 and 4 only
B. 1, 2, 4 and 5 only
C. 2, 3 and 5 only
D. 1, 3, 4 and 5 only

Answer: A

Solution:

Statement 1: Correct – Since both processing and budget constraints are binding, the optimal solution must lie at their intersection (corner point).

Statement 2: Correct – Finishing time has slack → not binding → does not affect optimal decision at optimum.

Statement 3: Incorrect – Removing demand constraint ($x \leq 5$) will shift optimum only if it was binding earlier. No indication that it was binding → cannot conclude "definitely."

Statement 4: Correct – Integer restriction may eliminate fractional optimal point → next feasible integer point yields lower contribution.

Statement 5: Incorrect – Even if budget increases, another constraint (processing time) is already binding → unless that is relaxed, output cannot increase → profit may remain unchanged.

2. A firm manufactures two products, X and Y, using three constrained resources. A Linear Programming model is formulated and solved using the Simplex Method. After multiple iterations, the final (optimal) tableau has the following properties:

- Both decision variables X and Y are basic variables in the final solution
- One slack variable has a positive value, while the remaining slack variables are zero
- All entries in the Z-row satisfy optimality conditions for a maximisation problem
- During one intermediate iteration, a pivot column had multiple identical minimum positive ratios
- In another iteration, one coefficient in the pivot column corresponding to a constraint was zero

Based on the above situation, which of the following statements are correct?

1. The presence of a positive slack variable in the optimal solution indicates that the corresponding constraint is non-binding.
2. If two or more equal minimum positive ratios occur during the ratio test, it may lead to degeneracy and potential cycling in subsequent iterations.
3. A zero coefficient in the pivot column for a constraint automatically disqualifies that row from being considered in the ratio test.
4. If both decision variables are basic in the optimal solution, it implies that all constraints must be binding.
5. For a maximisation problem, optimality is achieved when all reduced costs (Z-row coefficients excluding RHS) are non-negative.

A. Only 1, 2 and 5 are correct
B. Only 1, 3 and 4 are correct
C. Only 2, 3 and 5 are correct
D. Only 1, 2, 3 and 5 are correct

Answer: D

Solution:

Statement 1 – Correct: A positive slack variable indicates unused resource capacity. Hence, the constraint is non-binding at the optimum.

Statement 2 – Correct: Equal minimum ratios create degeneracy, where more than one leaving variable is possible. This can lead to cycling, especially in pathological cases.

Statement 3 – Correct: If a coefficient in the pivot column is zero, division for ratio test is not possible. Hence, that row is excluded from ratio consideration.

Statement 4 – Incorrect: Even if both decision variables are basic, it does not imply all constraints are binding. Presence of a positive slack variable already indicates at least one non-binding constraint.

Statement 5 – Correct: For maximisation, optimality condition requires all reduced costs (Z-row coefficients) to be ≥ 0 (no negative values remain).

1. A foreign exchange trading desk of a commercial bank wants to evaluate two position-management strategies using simulation modelling before implementing them in the live trading environment. The objective is to analyse how different position rules affect profitability and risk exposure under uncertain exchange rate movements.

The bank purchases a foreign currency at ₹82 per USD and sells it at ₹83 per USD. If the bank carries excess currency at the end of the day, it must square off the position in the interbank market at ₹81.50 per USD, resulting in a potential loss.

Two strategies are considered:

Strategy A (Reactive Strategy): Daily purchase quantity equals the previous day's market demand for USD.

Strategy B (Stabilized Strategy): Daily purchase quantity equals the average demand observed over the past 10 trading days, estimated to be $2 million per day.

Historical demand distribution observed by the bank is:

Daily Demand ($ million)	Probability	Random Number Interval
1.5	0.10	00–09
1.8	0.25	10–34
2.0	0.30	35–64
2.3	0.25	65–89
2.7	0.10	90–99

Using random numbers, the treasury department simulates exchange demand for 30 trading days to estimate profitability under both strategies.

Simulation results show: Average simulated demand: $2.05 million per day; Average purchase under Strategy A: $2.18 million; Average purchase under Strategy B: $2.00 million; Average executed sales under Strategy A: $1.92 million; Average executed sales under Strategy B: $1.97 million; Average daily profit under Strategy A: ₹1.68 million; Average daily profit under Strategy B: ₹2.21 million

The treasury committee reviews the findings before deciding the appropriate policy.

With reference to the above simulation exercise, consider the following statements:

1. The higher profitability of Strategy B primarily arises because the strategy aligns purchase quantities closer to the expected demand level, thereby reducing excess currency positions.
2. Strategy A leads to higher variability in purchase quantities because its ordering decision depends on realized demand of the immediately preceding trading day.
3. If the bank were allowed to carry forward excess currency without mandatory end-of-day square-off, the profitability difference between the two strategies would likely reduce.
4. Simulation modelling is particularly appropriate here because analytical optimization becomes difficult when demand uncertainty and operational constraints interact simultaneously.
5. The probability distribution used in the simulation guarantees that simulated demand will always converge exactly to the expected demand of $2 million over finite simulation runs.

A. 1, 2 and 4 only

B. 1, 2, 3 and 4 only

C. 2, 3, 4 and 5 only

D. 1, 2, 3, 4 and 5

Answer: B

Solution:

Statement 1 – Correct: Strategy B uses the mean demand estimate ($2 million). This stabilizes purchase decisions and reduces over-purchasing. Lower excess positions mean fewer square-off losses, improving profitability.

Statement 2 – Correct: Under Strategy A: $QA=Dn-1$. Since each day’s order depends on the previous day's realized demand, fluctuations in market demand directly transmit to purchase decisions, creating higher variability.

Statement 3 – Correct: The loss occurs because excess currency must be squared off at ₹81.50, creating a ₹0.50 loss per USD. If positions could be carried forward, the cost of over-purchasing would fall. This would reduce the penalty associated with Strategy A and therefore narrow the profitability gap between the two strategies.

Statement 4 – Correct

Simulation is useful when:

- System behaviour depends on random variables
- Multiple operational rules interact
- Analytical solutions become complex

Foreign exchange trading environments involve stochastic demand and operational constraints, making simulation appropriate.

Statement 5 – Incorrect: Probability distributions only ensure long-run convergence. Over a finite simulation period (30 days), the simulated demand may deviate from the expected mean due to randomness. Hence convergence to exactly $2 million is not guaranteed.

2. A retail distribution company plans to study its inventory ordering strategy using a simulation model. The objective is to maximise profits by selecting an optimal ordering rule. The decision maker can control the ordering rule, but daily demand fluctuates randomly and cannot be controlled. While designing the simulation, the analyst specifies parameters such as probability distributions of demand, decision rules, and a time-incrementing procedure.

The analyst considers two alternative time approaches:

- Fixed time increments, where the simulation clock advances daily and the system is checked at every time unit to determine whether an event occurs.
- Variable time increments, where the clock advances directly to the time of the next event, such as when inventory falls to a reorder level.

To ensure reliable insights, the analyst must also decide run length, choose statistical tests for evaluating results, and determine whether the model should run for a fixed period, until sufficient samples are generated, or until equilibrium between simulated and historical data is reached.

During discussions, the following statements were made regarding the design and application of simulation methodology:

1. A simulation model represents system elements through arithmetic, analog, or logical processes that replicate the dynamic behaviour of the real system.
2. Under the fixed time increment method, the simulation clock advances only when an event occurs in the system.
3. In the variable time increment method, time is advanced directly to the moment when the next event occurs, rather than scanning the system at every time unit.
4. Simulation models are generally reusable across multiple decision problems with minimal modifications, similar to linear programming models.
5. One method of determining simulation run length is to continue the simulation until simulated results stabilise and resemble historical patterns, indicating equilibrium.

Which of the following options is correct?

A. 1, 3 and 5 only

B. 1, 2, 3 and 5 only

C. 1, 3, 4 and 5 only

D. 1, 2, 4 and 5 only

Answer: A

Solution:

Statement 1 – Correct: Simulation models imitate real systems by representing system elements through arithmetic, analog or logical processes. These processes allow the model to reproduce the dynamic behaviour of the actual system and predict its performance under different conditions.

Statement 2 – Incorrect: Under fixed time increments, the simulation clock advances at regular intervals (e.g., day by day or hour by hour). At each time step, the system is scanned to check whether any event occurs. Even if no event occurs, the clock still advances by one time unit. Therefore, time does not depend on event occurrence.

Statement 3 – Correct: In the variable time increment method, the simulation clock jumps directly to the time of the next event. This method is efficient when events occur irregularly or infrequently, such as when an order is triggered only when inventory reaches a reorder level.

Statement 4 – Incorrect: Simulation models are custom-built for specific problems. Unlike models such as linear programming, which can be reused by simply modifying objective function coefficients and constraints, simulation models usually require substantial redesign for different systems.

Statement 5 – Correct: One approach for determining simulation run length is to continue running the model until equilibrium is reached, meaning simulated data begins to resemble historical or expected patterns.

Other approaches include:

- Running the simulation for a specified time period
- Running it long enough to generate large samples

UNIT 9

Simulation

STRUCTURE

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B. 1, 2, 3 and 5 only

C. 1, 3, 4 and 5 only

D. 1, 2, 4 and 5 only

Answer: A

Solution:

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Statement 5 – Correct: One approach for determining simulation run length is to continue running the model until equilibrium is reached, meaning simulated data begins to resemble historical or expected patterns.

Other approaches include:

- Running the simulation for a specified time period
- Running it long enough to generate large samples

3. If a sample of 200 batteries has a mean life of 36 months and a standard deviation of 10 months, what is the **standard error of the mean**?

A. 0.507

B. 0.707

C. 0.807

D. 0.907

$$\frac{10}{\sqrt{200}}$$

Answer: B

The formula for **standard error of the mean** is:

SE= s/Vn = 10/V200 - 10/14.14 - 0.707 So, the correct answer is **0.707 months**

4. Which of the following standard error multipliers corresponds to a 99% confidence interval?

A. ±1.64 90

B. ±1.96 95

C. ±2.58 99

D. ±2.33 98

Answer: C

For a **99% confidence level**, the standard error range used is **±2.58**, which covers approximately **99% of the area under a normal curve**.

- ±1.64 covers about 90%
- ±1.96 is typically used for 95%
- ±2.33 corresponds to roughly 98%

5. If a sample of 75 employees shows that 40% prefer their own retirement benefits, what is the estimated standard error of the proportion?

A. 0.057
B. 0.075
C. 0.040
D. 0.600

$$\sqrt{\frac{.4 \times .6}{75}}$$

Answer: A

The estimated standard error of the sample proportion is calculated using the formula:

$op = pq/n = V0.4 \times 0.6/ 75 = 0.0032 - 0.057$

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